Modernizing a 2M+ LOC Java Monolith with Agentic AI

What Worked, What Failed, and How We Recovered

Marek Dominiak



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Agenda



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- The modernization problem



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- Agentic AI in practice



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- V2 meets production



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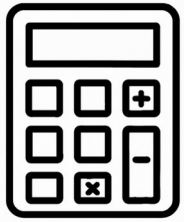
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- Q/A



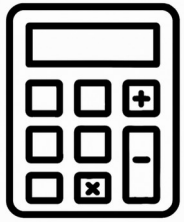
The modernization problem

The system we had to modernize



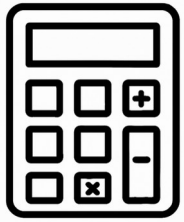
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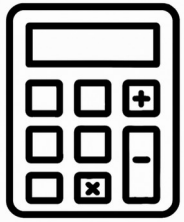
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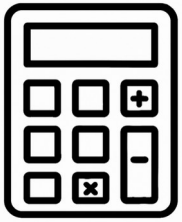
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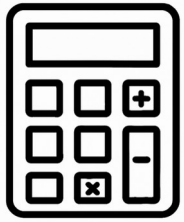
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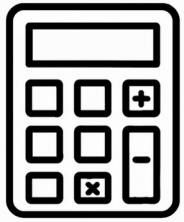
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- Lines of code: > 2M
- Number of user accounts: ~ 3M



Why did we need to upgrade?



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- Worsened Developer Experience



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- Increased cost of keeping things secure

Max CVSS	9.8
EPSS Score	49.26%
Published	2018-01-04
Updated	2022-04-07

Max CVSS	9.8
EPSS Score	0.74%
Published	2023-04-20
Updated	2023-08-28

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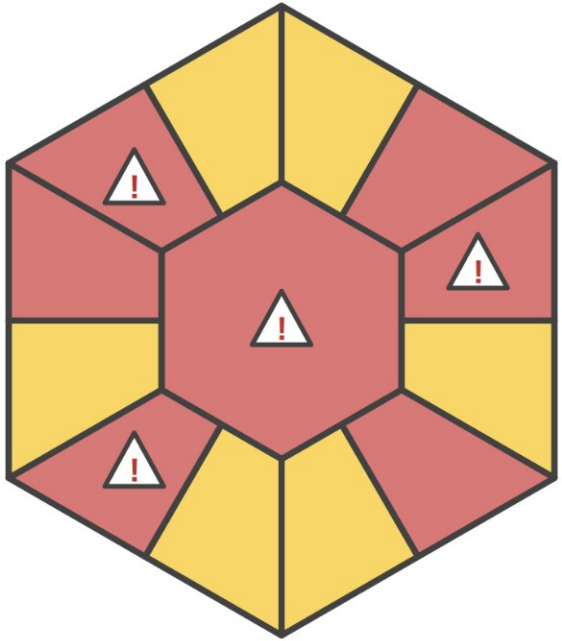


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- Increased costs related to running outdated versions on Cloud (e.g. Mysql on AWS)

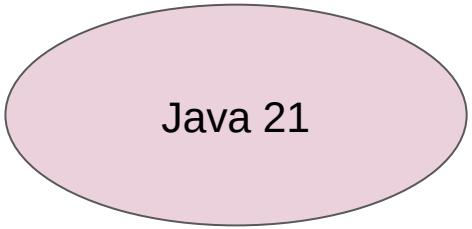




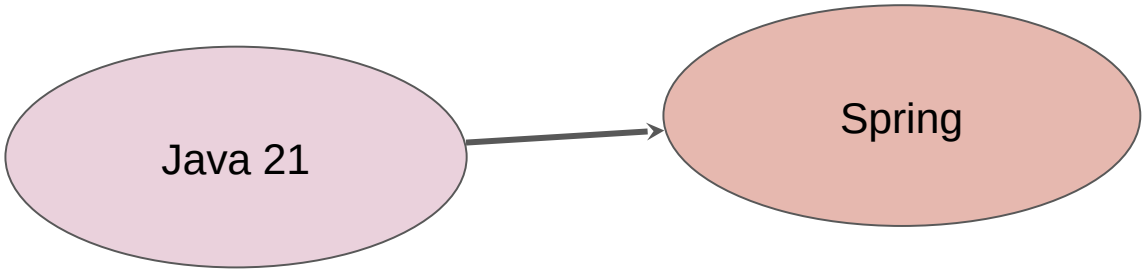
Before: one monolith, many parts, many of them with issues.

Chained Dependencies

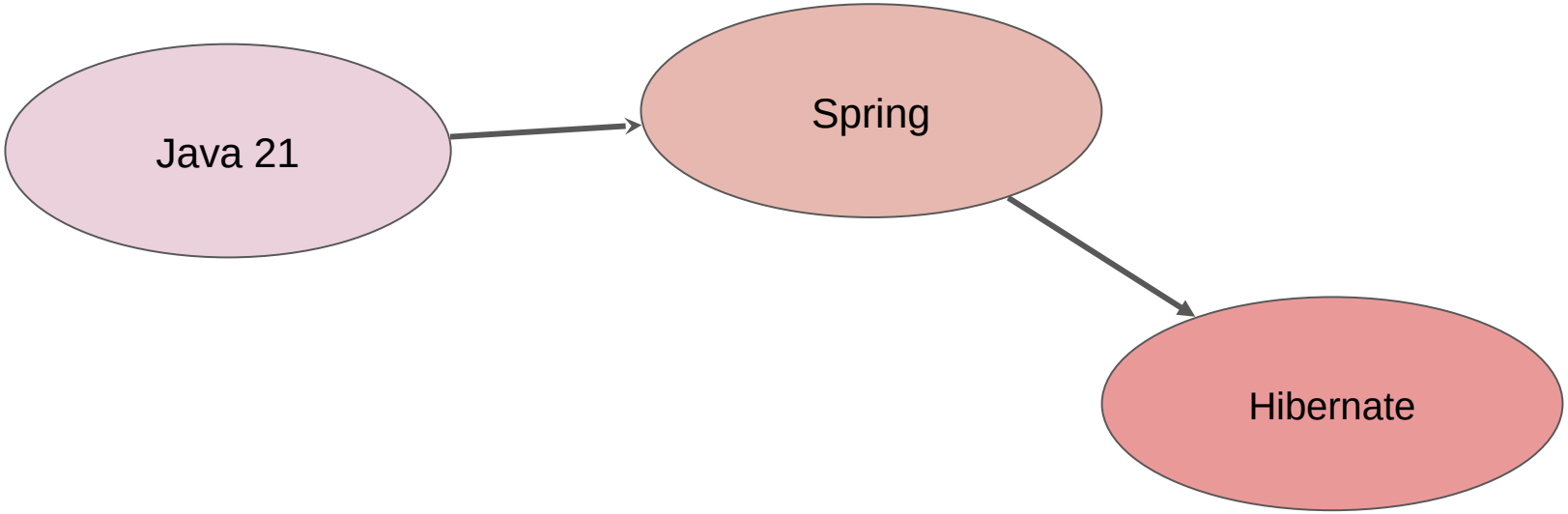
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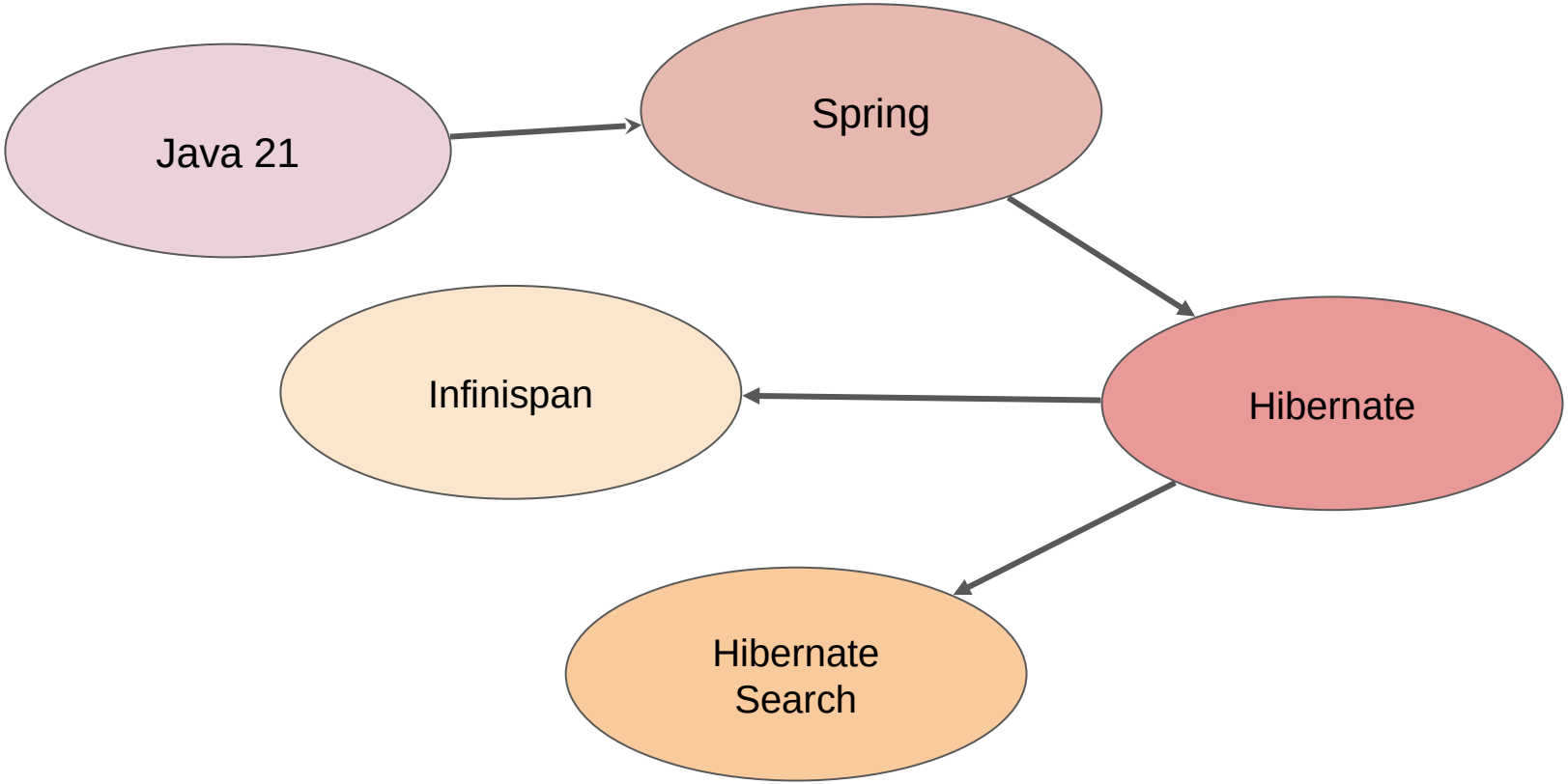
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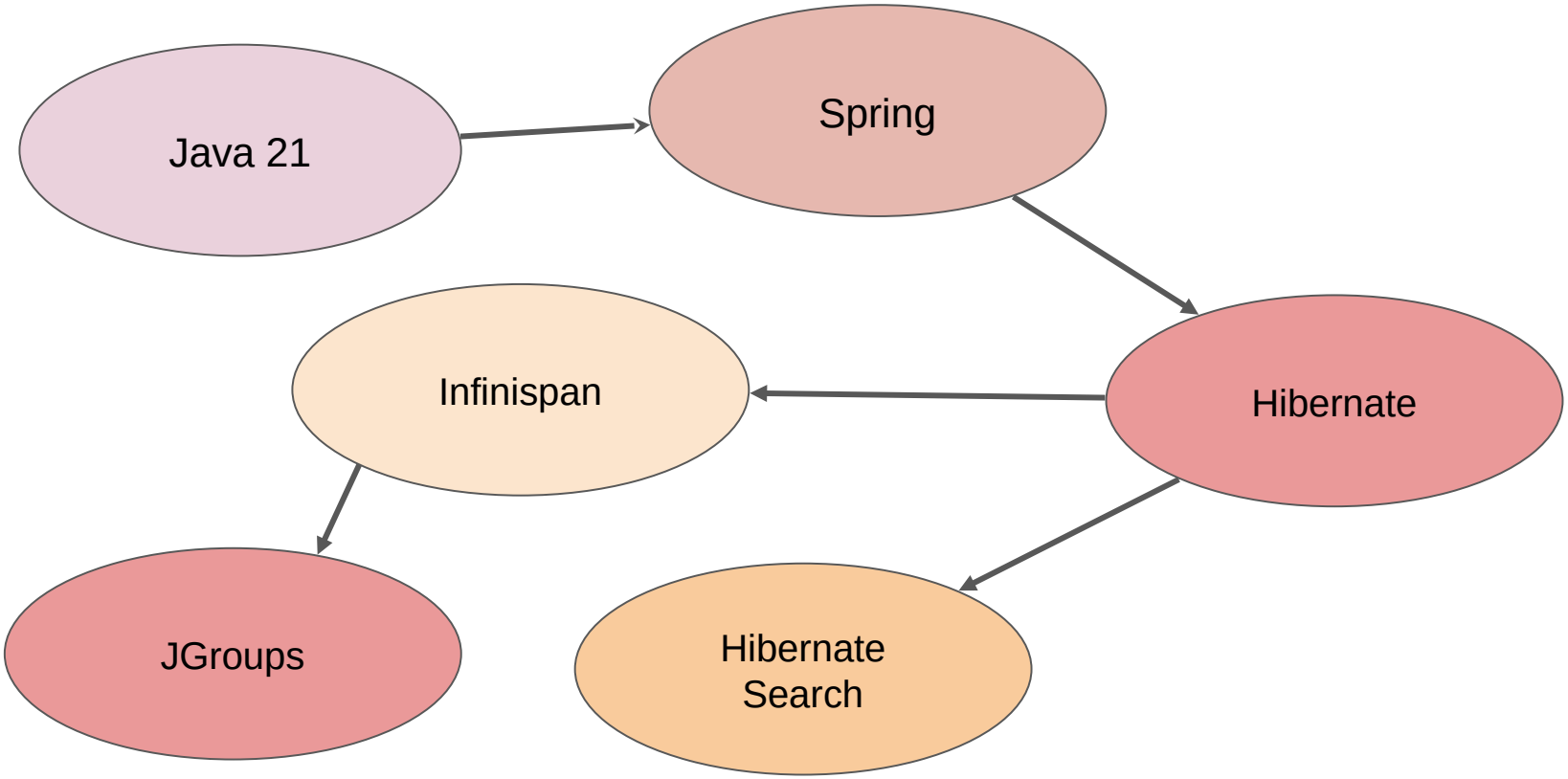
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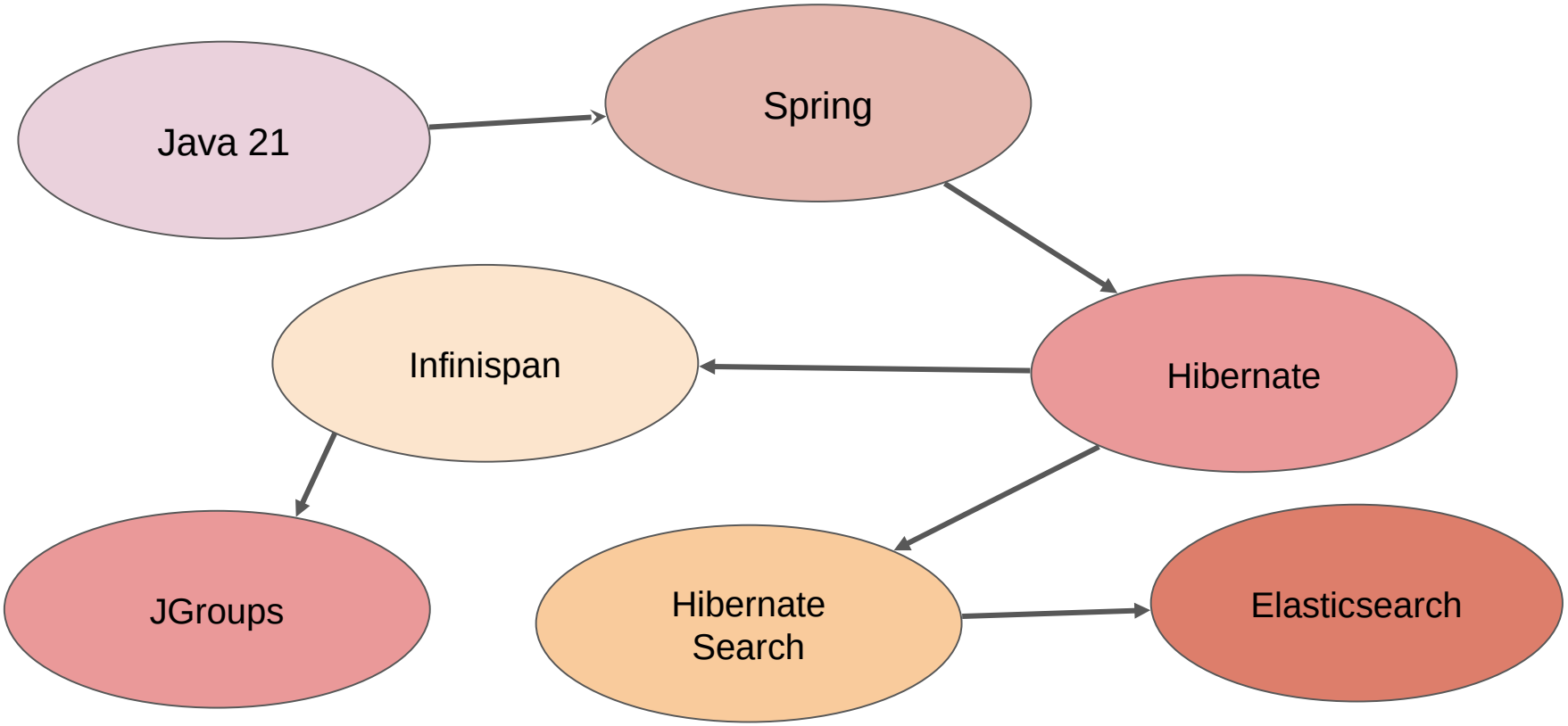
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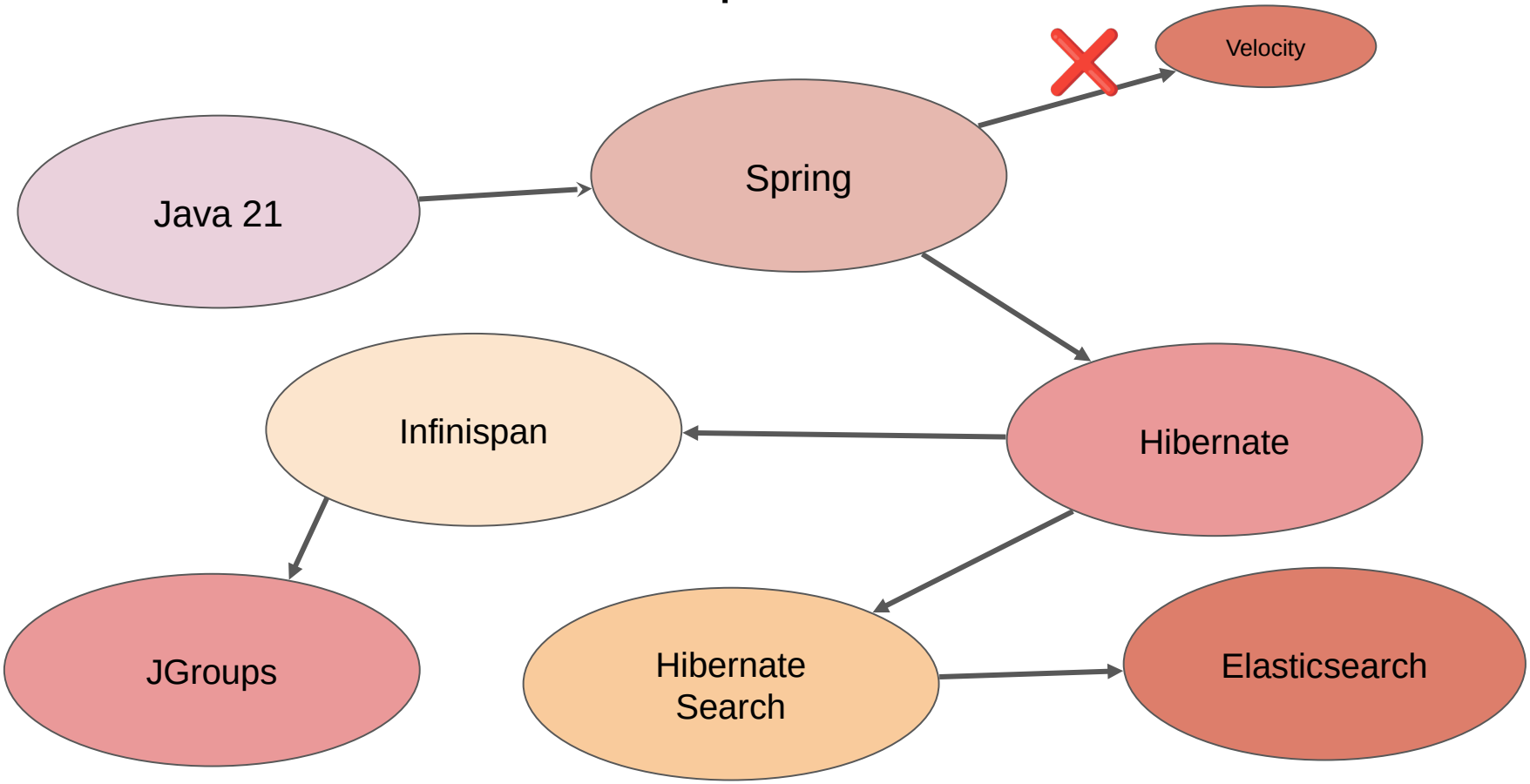
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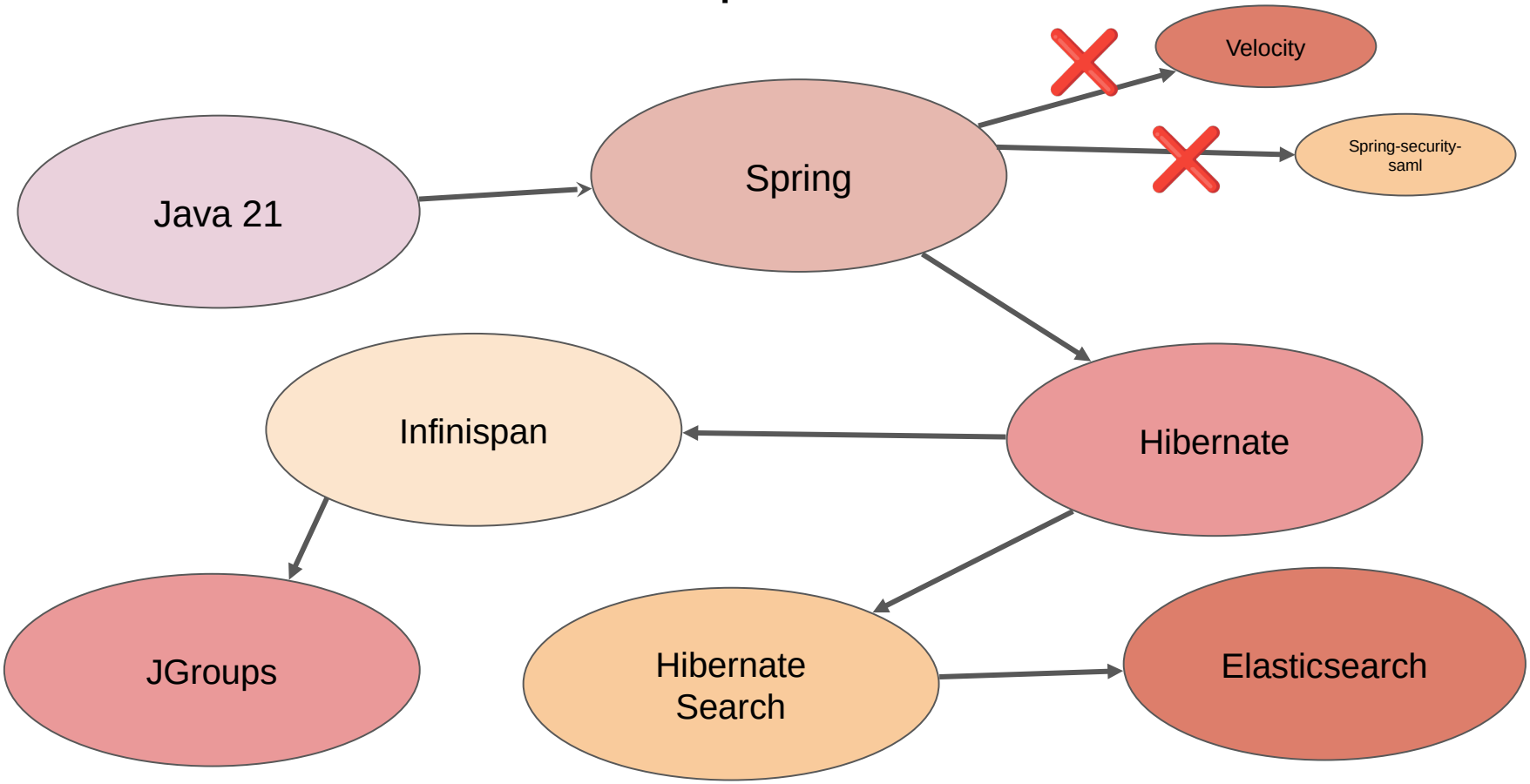
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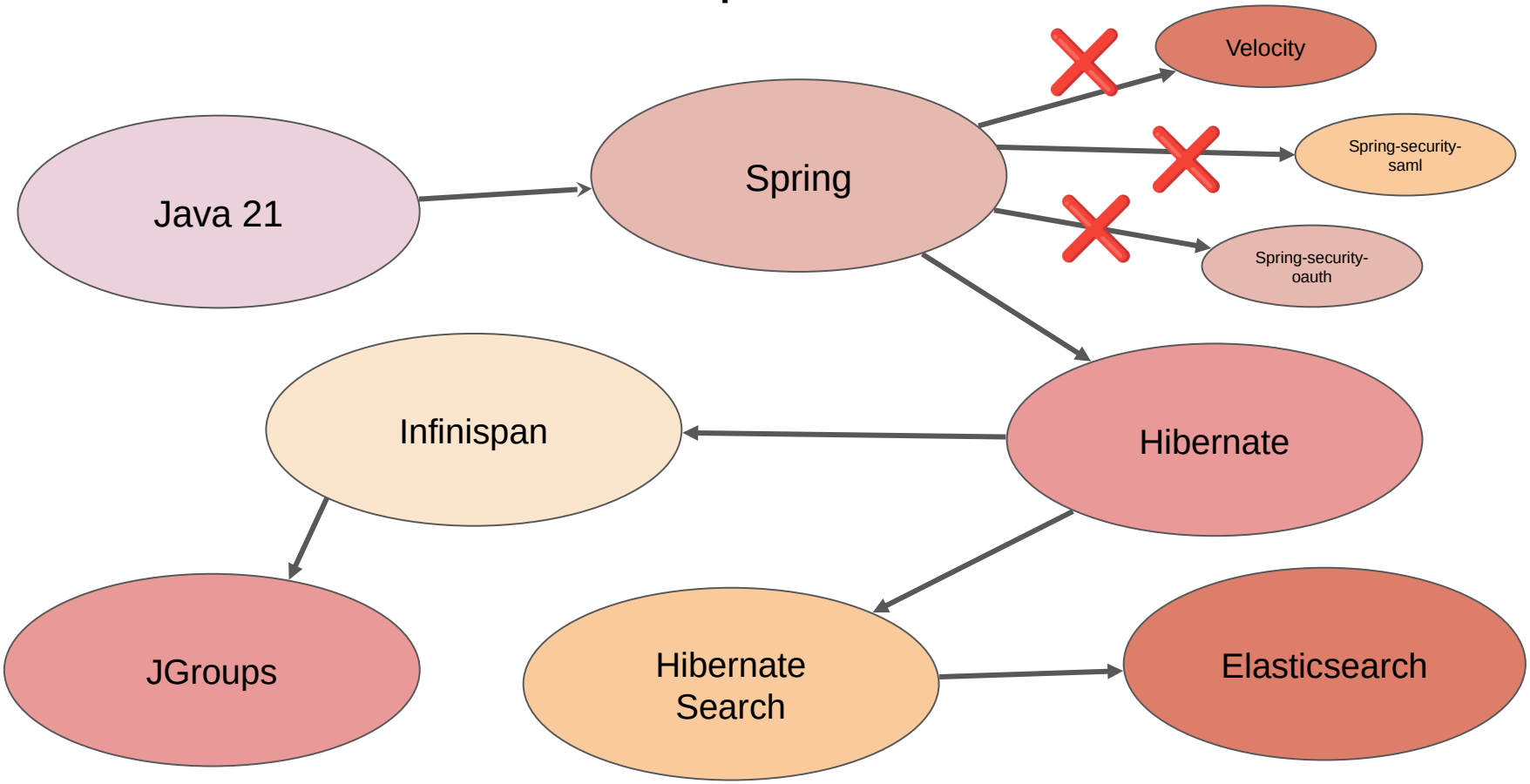
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Our strategy: an upgrade branch with weekly merges

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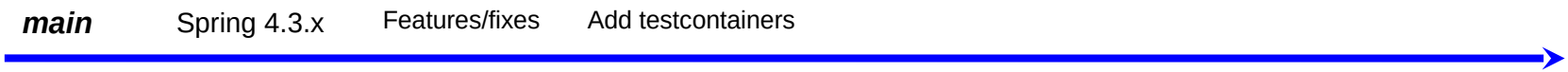
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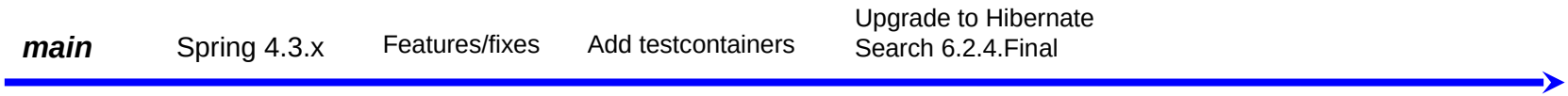
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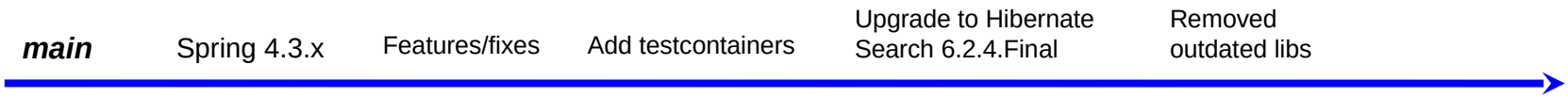
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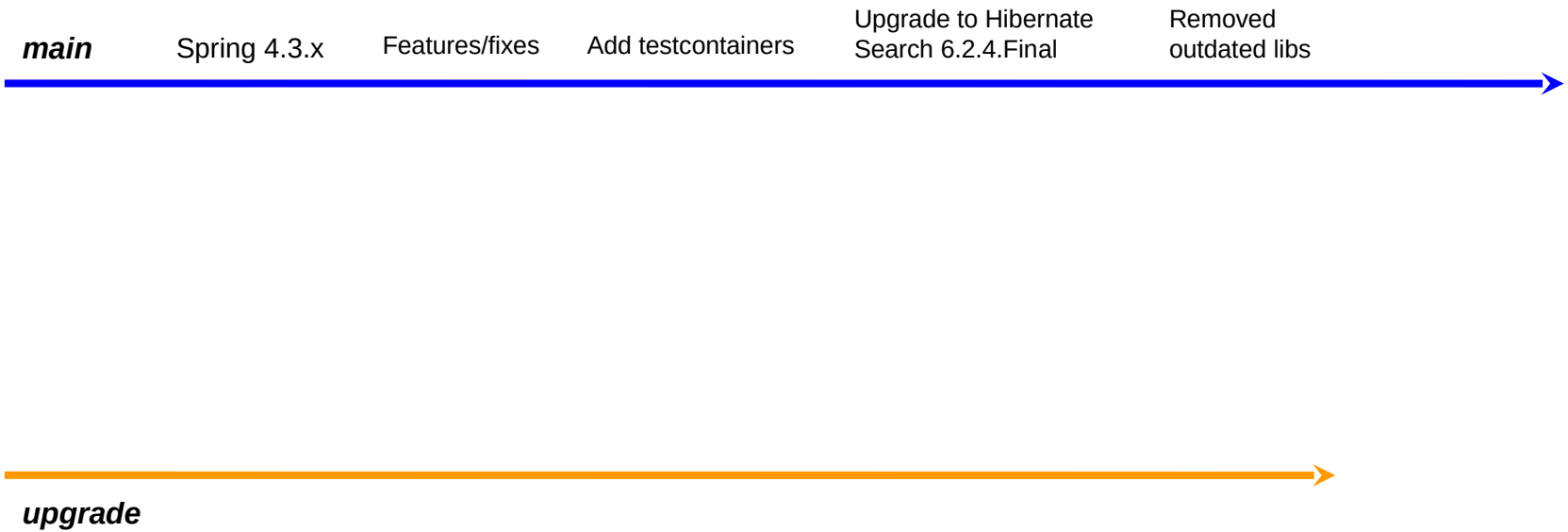
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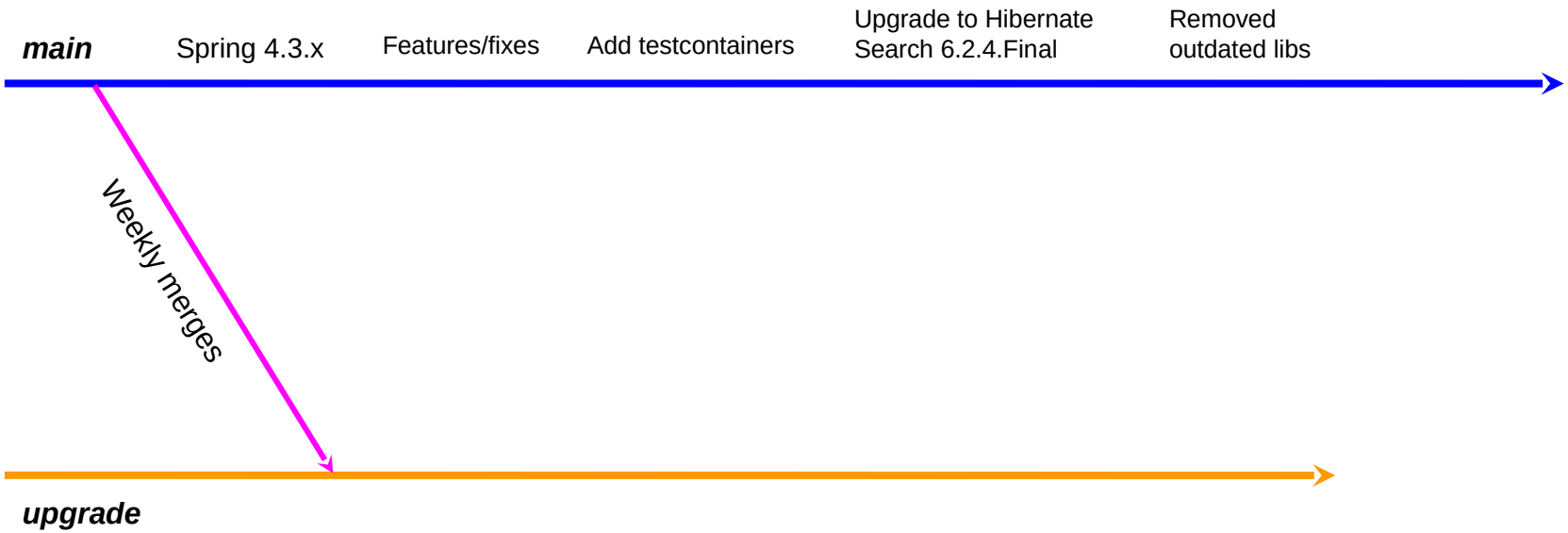
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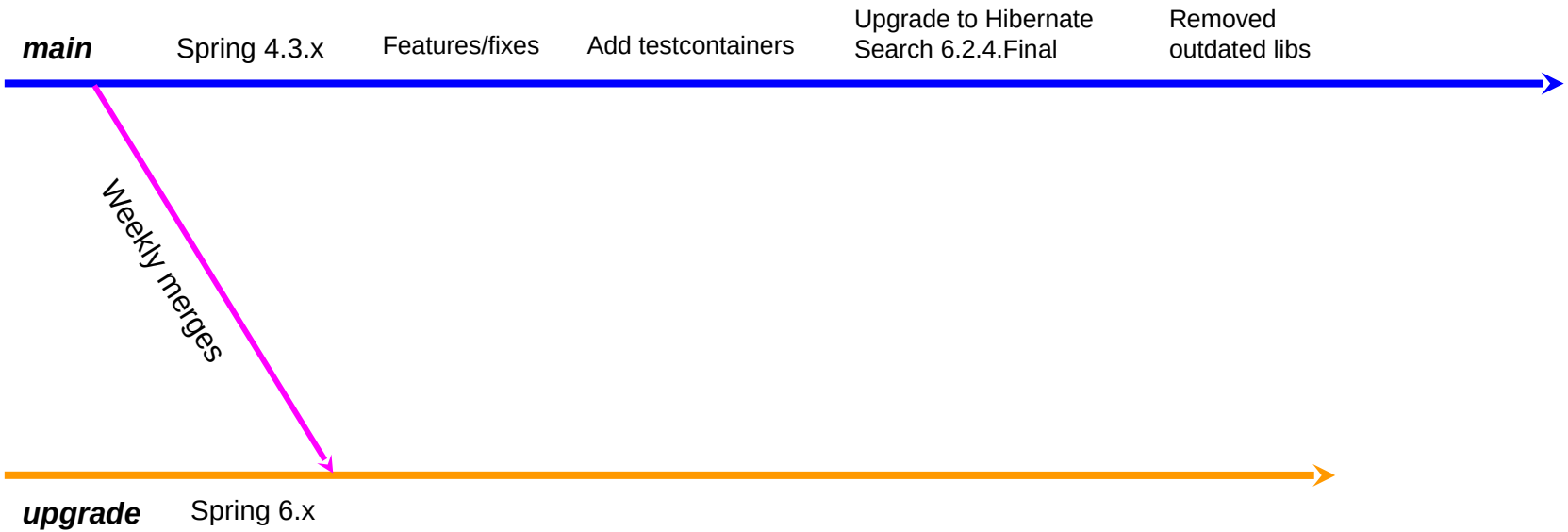
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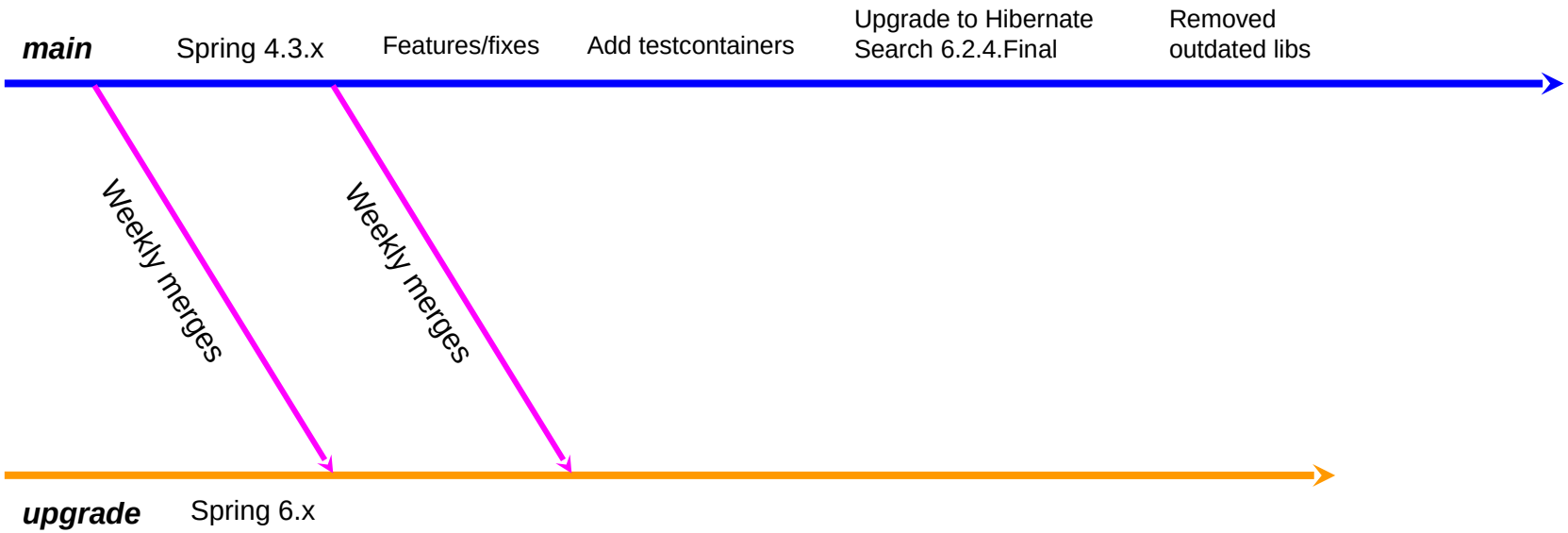
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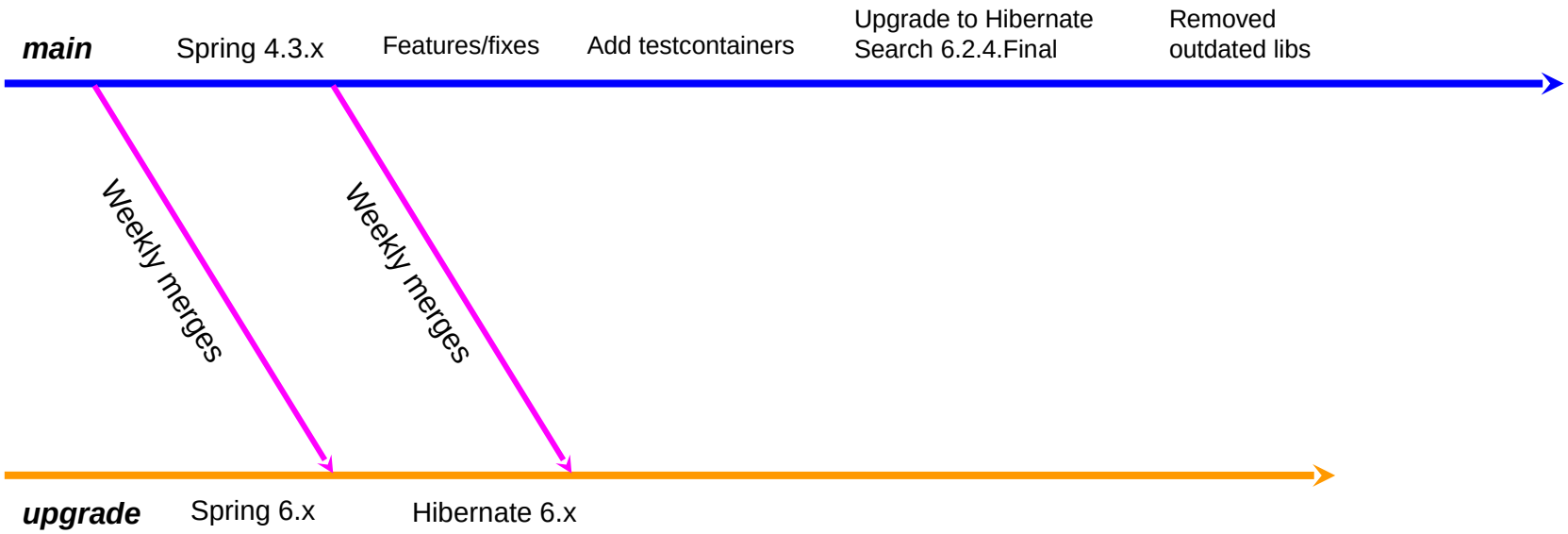
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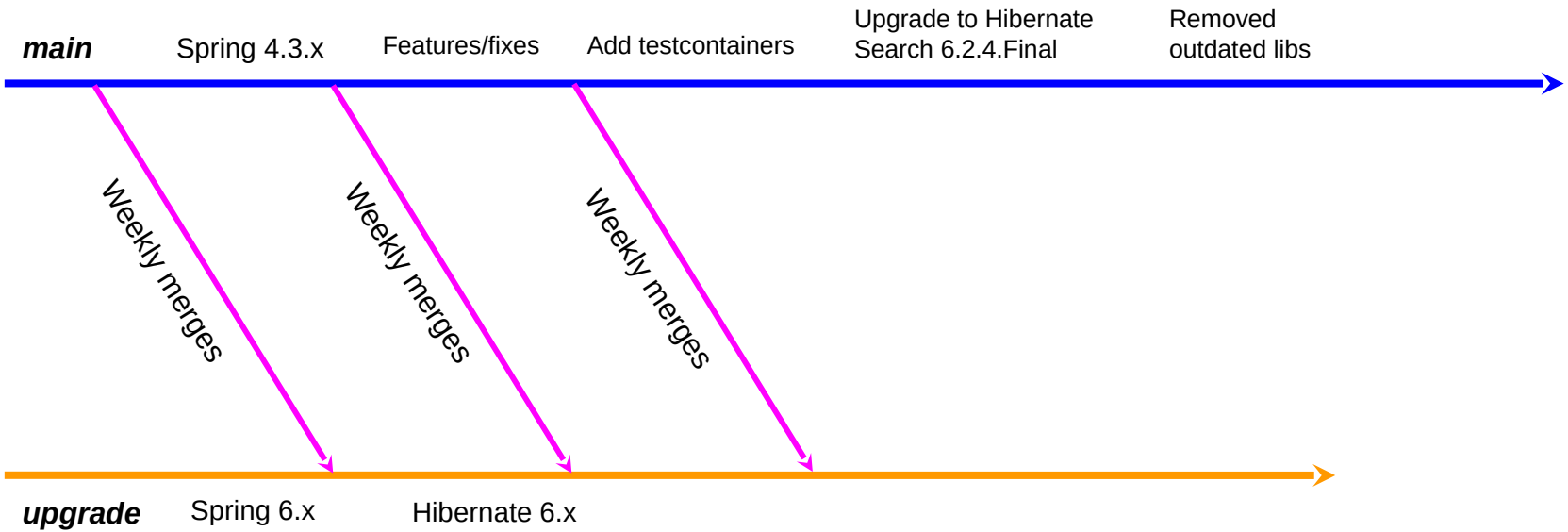
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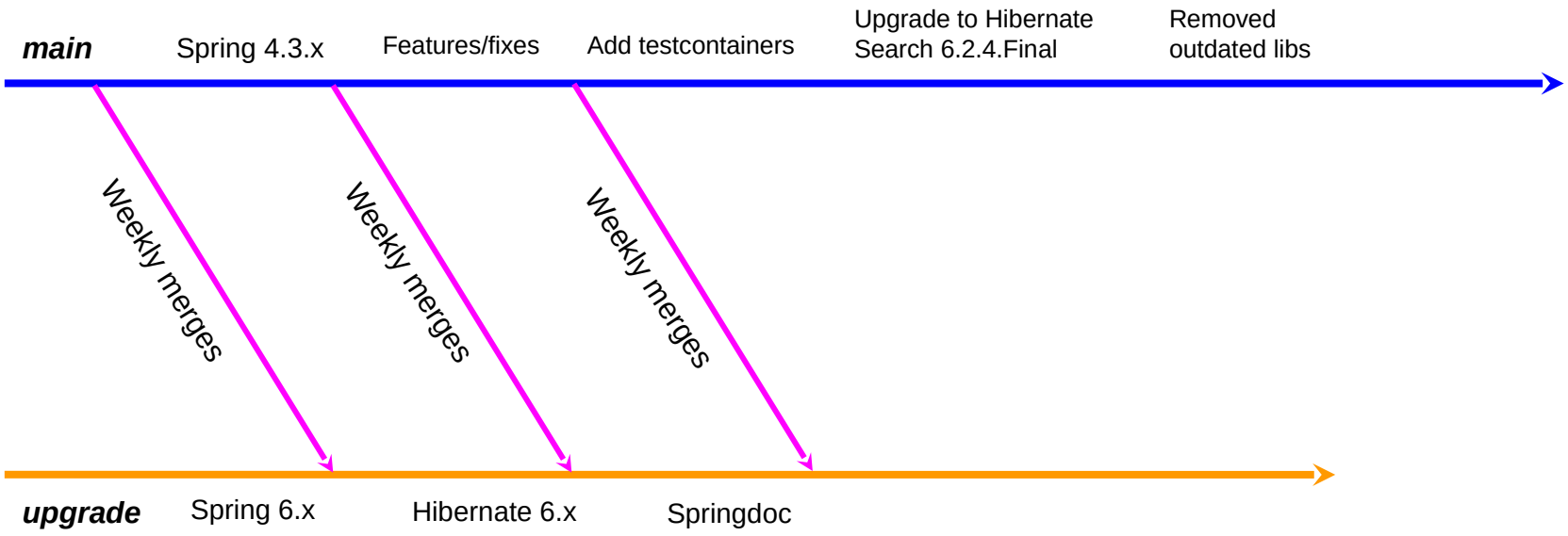
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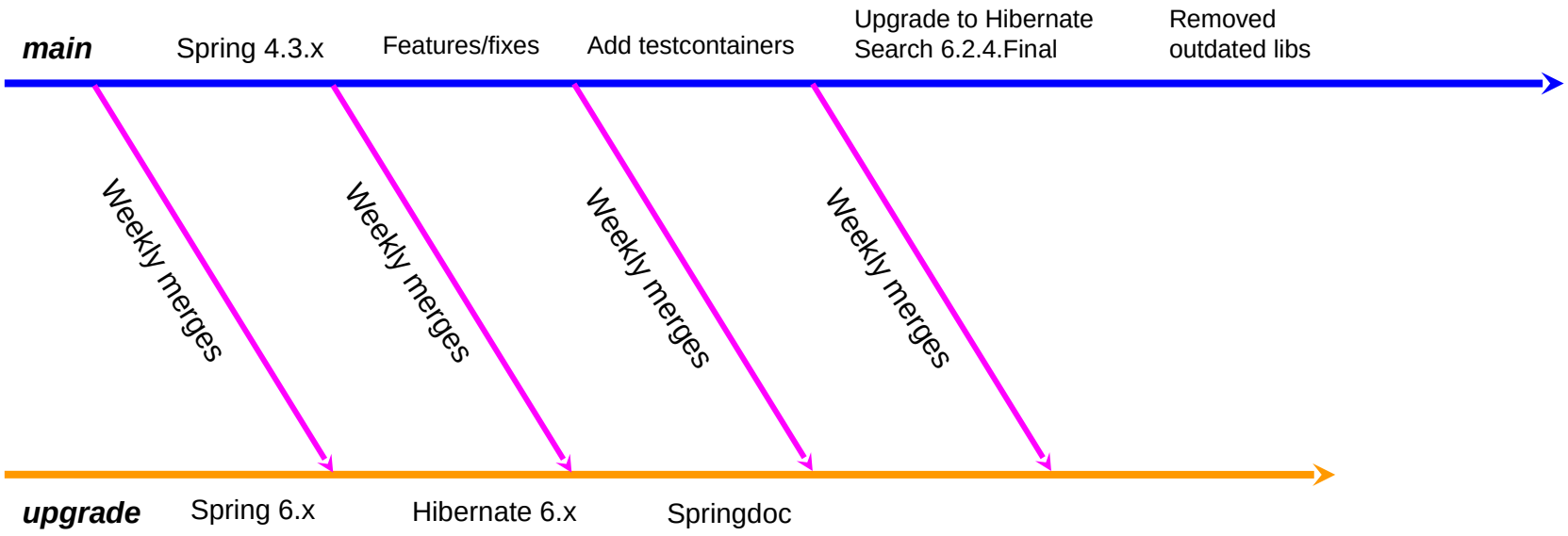
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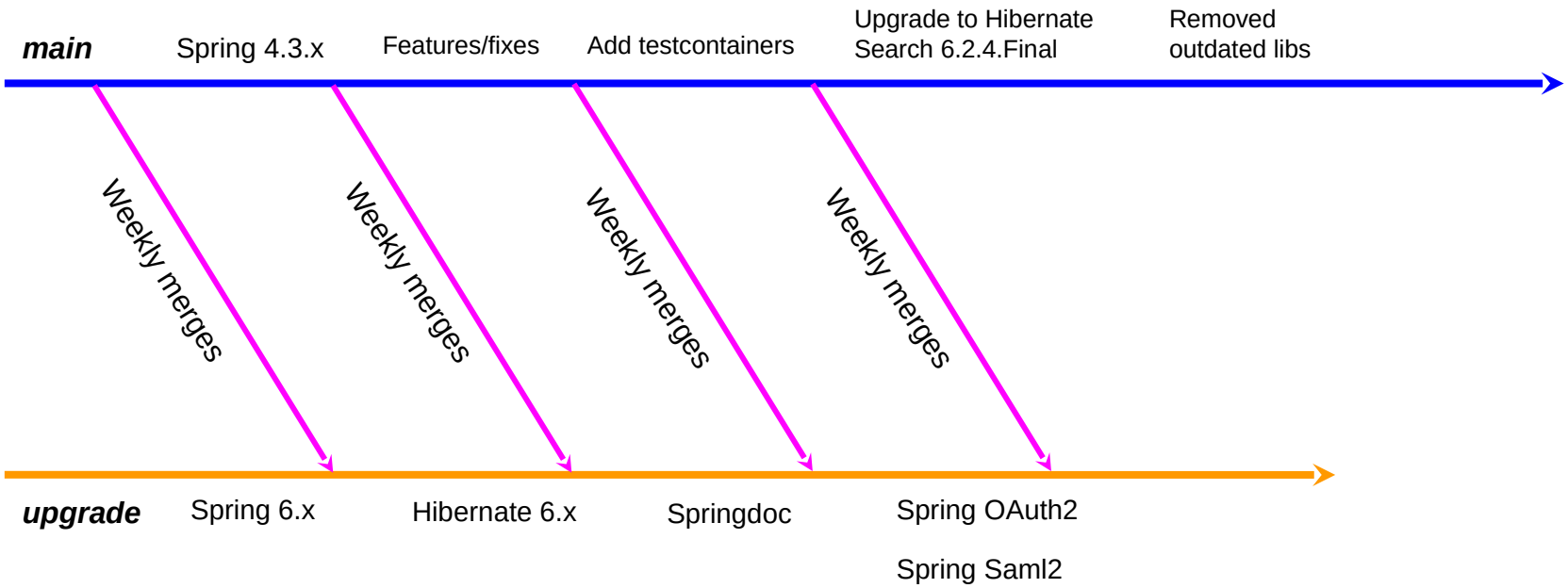
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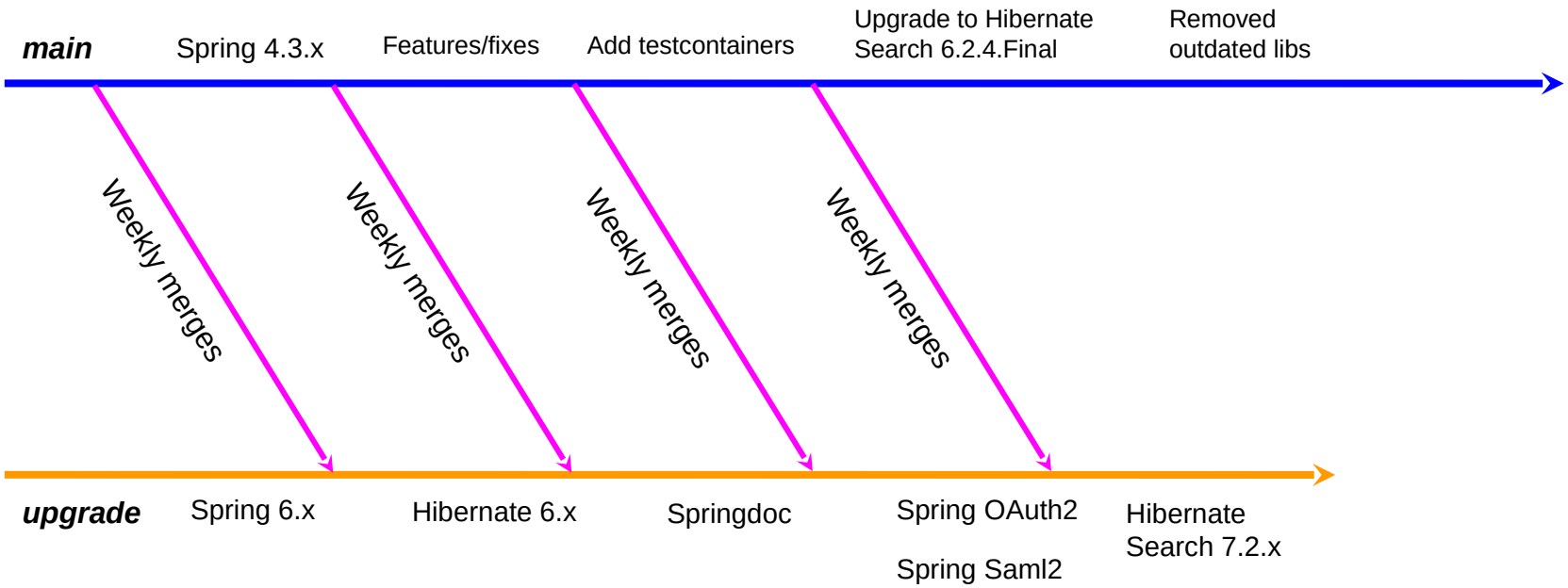
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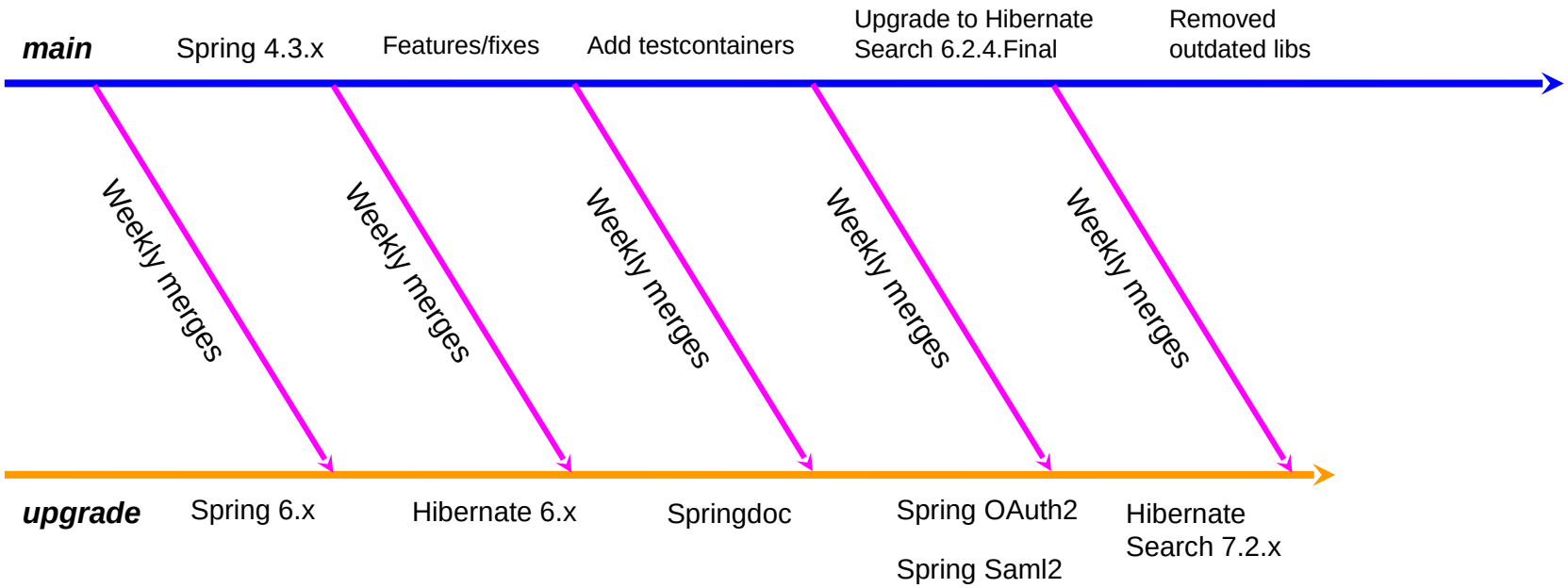
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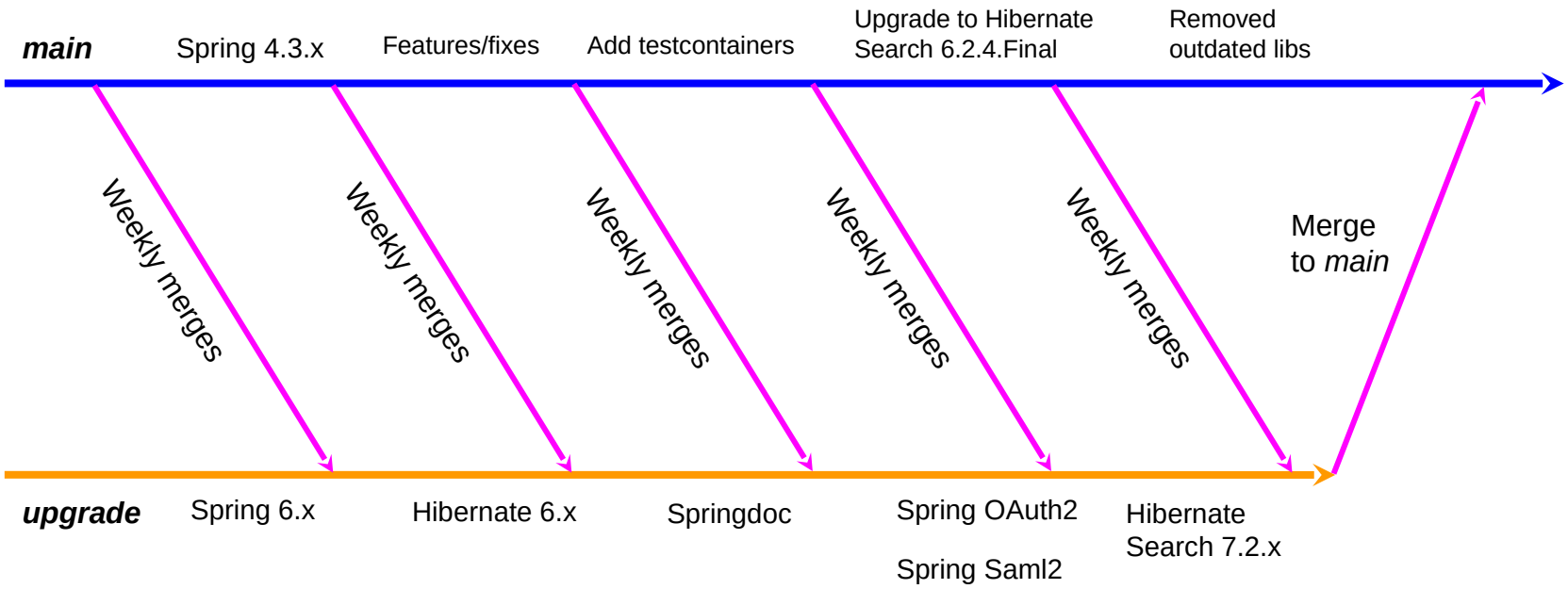
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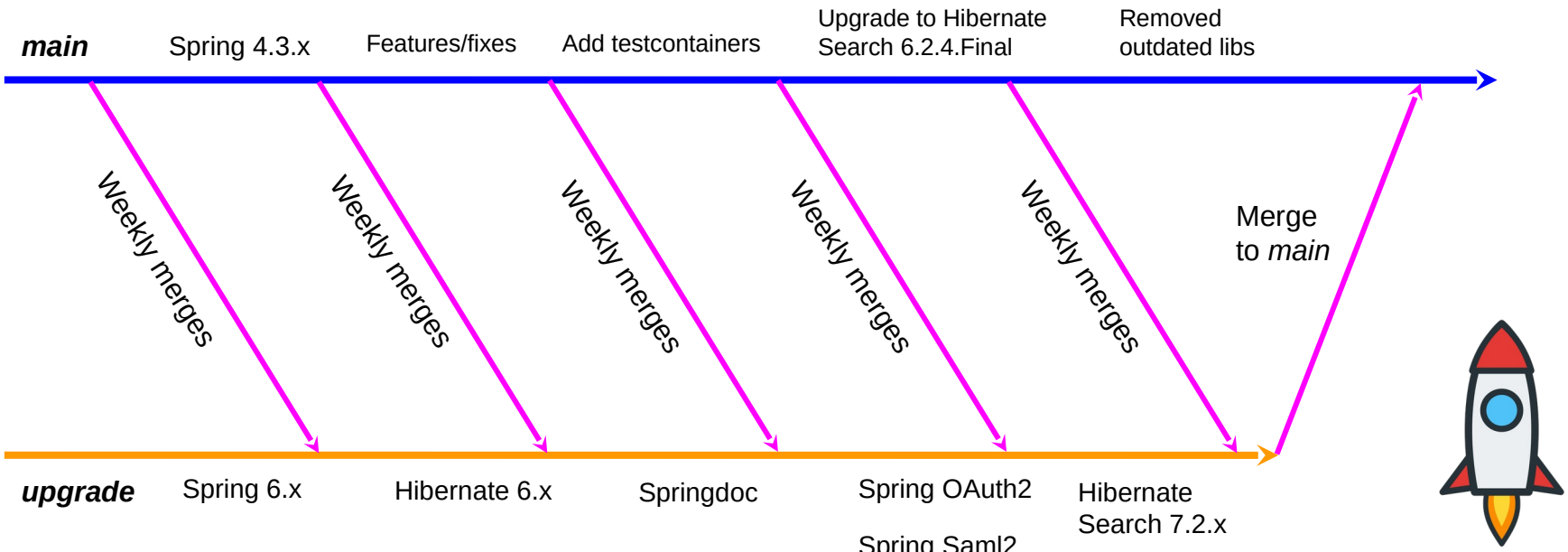
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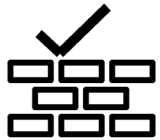
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Agentic AI in practice

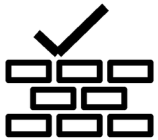
Our starting point



Agents made us faster because these were already in place.

Our starting point

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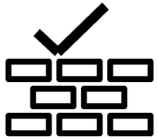


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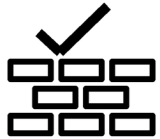


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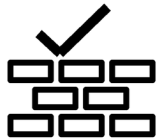


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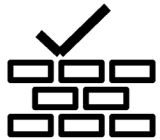


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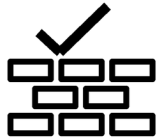


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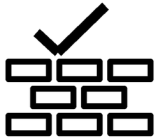
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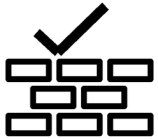
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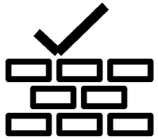
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- Enough people for the modernization
- Time, next to new features and integrations

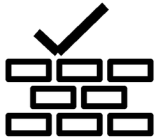
Agents made us faster because these were already in place.

Our starting point

What we had

- **Tests we trust:** unit and integration, Testcontainers
- **CI** on every merge request
- **ADRs** for important decisions
- **Monitoring** in production
- **A team** that knows the domain and the code

What we didn’t have



- Enough people for the modernization
- Time, next to new features and integrations
- A clear idea of how long it would take

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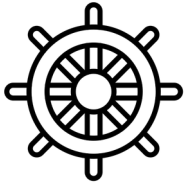
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My merged MRs, 4 weeks of work

Merged !2753

Remove dead code and unused test fixtures

106 files **+85** **−1,575,684** *mostly CSV test fixtures*

Merged !2754 → feature/tp-stack-upgrade

Port master to Java 21 / Spring 6 / Hibernate 6

1,553 files **+64,427** **−34,456** *166 conflicts*

Merged !2678

Remove obsolete setup scripts and config profiles

193 files **+184** **−19,126**

Merged !2626

Empty local database from the production schema

9 files **+18,969** **−3** *mostly generated schema*

Merged !2748

Remove tests that give no value

20 files **+3** **−3,126**

Merged !2696

Replace S3Mock with the Floci S3 emulator

15 files **+259** **−1,555**

Merged !2737

Group JMS queues for several worker nodes

31 files **+465** **−139**

Merged !2736

Faster test MySQL and Elasticsearch

21 files **+200** **−48** *CI: 1 h 55 → 1 h 28*

22 MRs in total. Done in a fraction of my time. Without agents, not possible.

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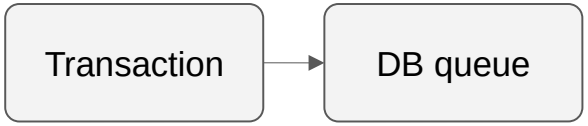
Too many moving parts, and we lost control of priorities.

How V1 worked

Transaction

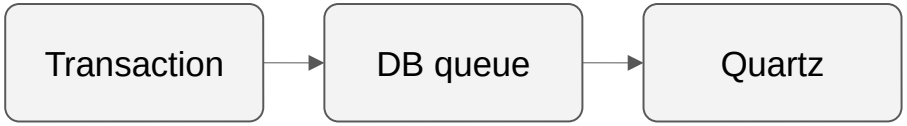
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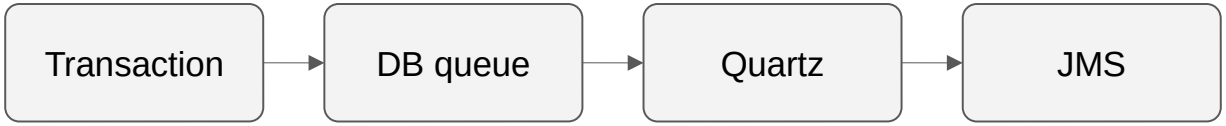
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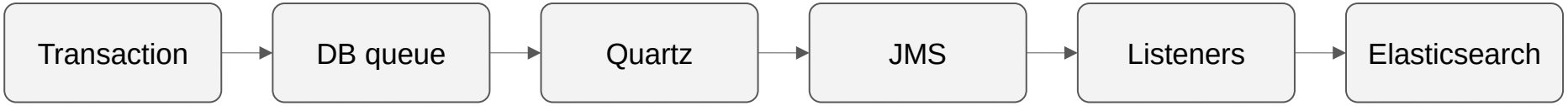
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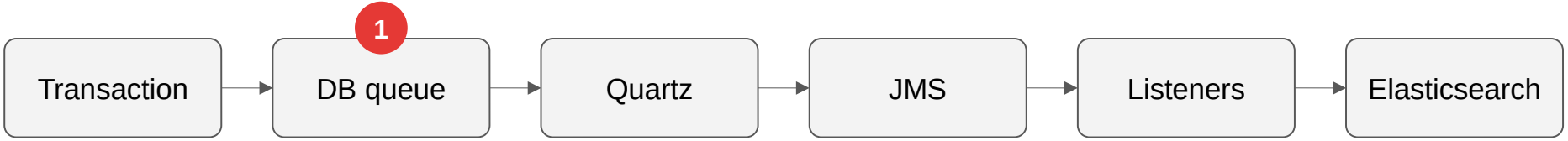
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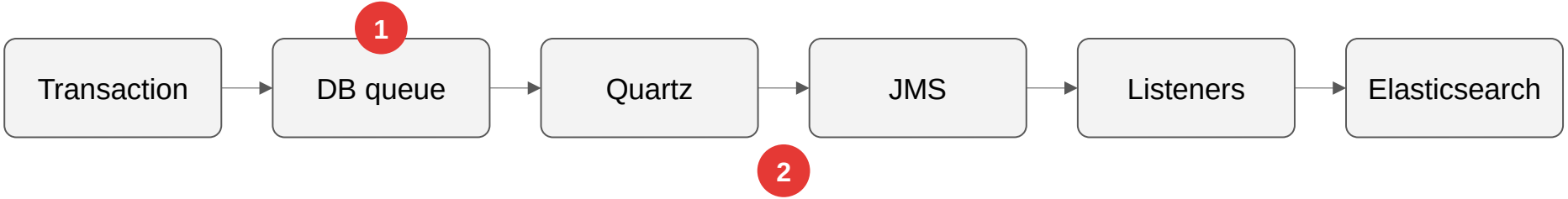
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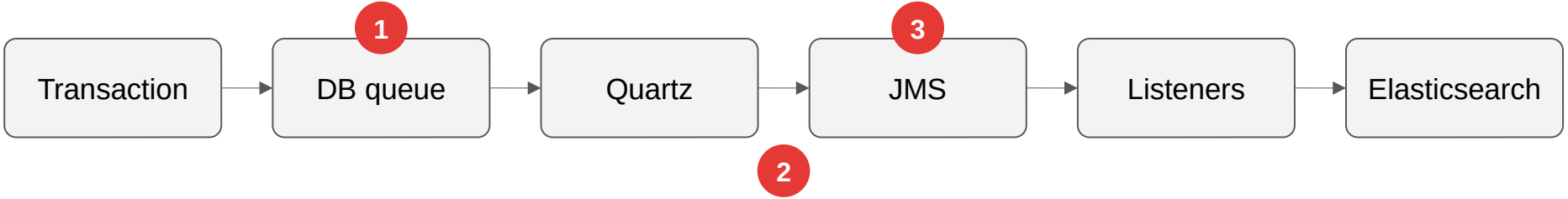
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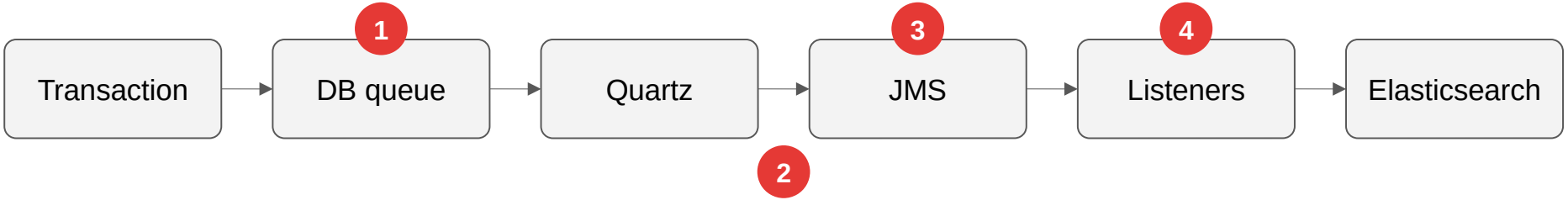
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4. **Scaling:** adding consumers needed a server restart

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TP-013 - Improve control of priorities, and performance of Indexing Queues

By Marek Dominiak 6 min Listen 18 Add a reaction

Status	APPROVED IMPLEMENTED
Impact	MEDIUM
Drivers	Maintainability, Flexibility, Fault Tolerance
Approver	Architecture board:
Due date	Production: Jul 29, 2026

What the ADR decided

Remove JMS. Let the database do the queueing.

What the ADR decided

Rejected

Option 1

Hibernate Search outbox polling

+ Out of the box, battle-tested
+ Deduplication per transaction
− Still incubating
− We lose priorities and fairness

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What the ADR decided

Rejected

Option 1

Hibernate Search outbox polling

+ Out of the box, battle-tested
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− Still incubating
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Chosen

Option 2

MySQL queue, no JMS

+ We stay in control of priorities and fairness
+ SELECT … FOR UPDATE SKIP LOCKED: many consumers, no contention
− We build polling and deduplication ourselves

Remove JMS. Let the database do the queueing.

V2 implementation

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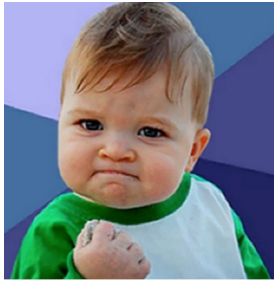
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Ref: https://ghuntley.com/loop/

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Step 50 — Preserve error state for permanently failed items on re-enqueue. 184 tests pass.	Marek Dominiak	Yesterday 12:59 PM
Step 49 — Error Rate Alarm. 182 tests pass.	Marek Dominiak	Yesterday 12:45 PM
Step 48 — Populate/Re-index All Items. 174 tests pass.	Marek Dominiak	Yesterday 12:21 PM
Step 47 — Replace wildcard imports in test files. 174 tests pass.	Marek Dominiak	Yesterday 7:07 AM
Step 46 — Replace wildcard imports with explicit imports. 174 tests pass.	Marek Dominiak	Yesterday 6:53 AM
Step 45 — Update docs and Postman for Steps 41-44. 174 tests pass.	Marek Dominiak	Yesterday 6:32 AM
Step 44 — Production-ready auto-scale defaults. 174 tests pass.	Marek Dominiak	Yesterday 6:25 AM
Step 43 — Remove expensive stale item COUNT queries. 174 tests pass.	Marek Dominiak	Yesterday 5:54 AM
Step 42 — Wall-clock batch time stats. 182 tests pass.	Marek Dominiak	Yesterday 5:40 AM
Step 41 — In-memory staleness check (zero DB queries). 182 tests pass. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 5:39 PM
Step 40 — Update docs and Postman collection. 182 tests pass. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 1:48 PM
Step 39 DRY review. 182 tests pass. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 1:33 PM
Step 37 — Consolidate contract-expand scripts. 211 tests pass. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 12:39 PM
Step 36 — Improve observability: commit/flush timing + staleness check throttling. (1) Commit/flush timing: Stopwatch wrapping transactionTemplate.execute() captures Hiber	Marek Dominiak	3/10/26, 12:15 PM
Step 35 — Remove autoTuneBatchSize feature (keep autoScaling of consumers). Removed batch size auto-tuning: fields/methods from IndexerProperties, IndexConsumer, Cor	Marek Dominiak	3/10/26, 11:44 AM
Step 34 — Include IndexConfiguration (batchSize) in properties REST endpoints. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 11:36 AM
Step 33 — Add comprehensive Javadoc to IndexerProperties and configuration classes. Co-Authored-By: Claude Opus 4.6 <noreply@anthropic.com>	Marek Dominiak	3/10/26, 11:28 AM
Step 32 - `EntityIndexQueueRepositoryImpl.java` — Replaced `fetchRemainingExcludingOverrides` (NOT IN query) with `fetchByEntityClassCount` (per-class queries mirroring	Marek Dominiak	3/9/26, 3:12 PM
Step 31 — Soft-delete-aware fetch performance indexes. Add processedAt as leading column to all composite fetch indexes so MySQL can skip soft-deleted rows via index se	Marek Dominiak	3/9/26, 2:05 PM
Step 30 - Eliminate per-entity-class fetch loop. Replace 25+ SELECT FOR UPDATE SKIP LOCKED queries per batch with single NOT IN query. Co-Authored-By: Claude Opus 4.6	Marek Dominiak	3/9/26, 1:53 PM
Step 29 — Entity fetch optimization + status config	Marek Dominiak	3/9/26, 12:23 PM
Step 28 — Documentation & Postman audit	Marek Dominiak	3/6/26, 1:41 PM
Step 27 — Developer FAQ: Group stats lifecycle	Marek Dominiak	3/6/26, 1:22 PM
Step 26 — REST stop-all / start-all endpoints	Marek Dominiak	3/6/26, 1:09 PM
Step 25 — Periodic stats reconciliation	Marek Dominiak	3/6/26, 12:49 PM
Step 24 — Default max consumers 10 → 3	Marek Dominiak	3/6/26, 12:20 PM
Step 23 — Plan for bullet-proofing indexing infrastructure	Marek Dominiak	3/6/26, 12:11 PM
Step 22 — Troubleshooting guide + solution comparison docs sync	Marek Dominiak	3/6/26, 6:51 AM
Step 21 — Reconcile group stats on coordinator startup	Marek Dominiak	3/6/26, 6:42 AM
Step 20 — ADR + Developer FAQ update after in-memory cache	Marek Dominiak	3/6/26, 6:24 AM
Step 19 — Tests for in-memory pendingDecrements cache	Marek Dominiak	3/6/26, 6:12 AM
Step 18 — Wire flushPendingDecrements into IndexCoordinator lifecycle	Marek Dominiak	3/6/26, 6:05 AM
Step 17 — In-memory pendingDecrements cache for decrementGroupStats	Marek Dominiak	3/6/26, 5:53 AM
Step 16 — Plan: In-memory group stats cache to eliminate decrementGroupStats bottleneck	Marek Dominiak	3/6/26, 5:25 AM

Keep the sessions small

V2 meets production

Around 250 Ralph loops later…

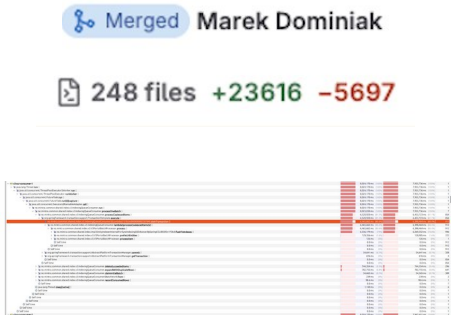
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Merged Marek Dominiak

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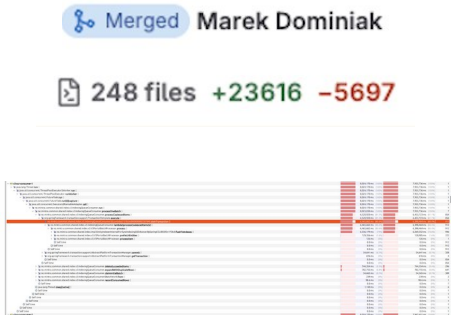


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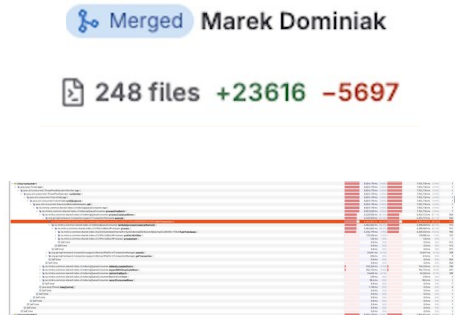
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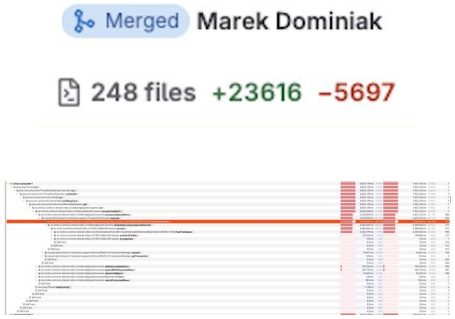
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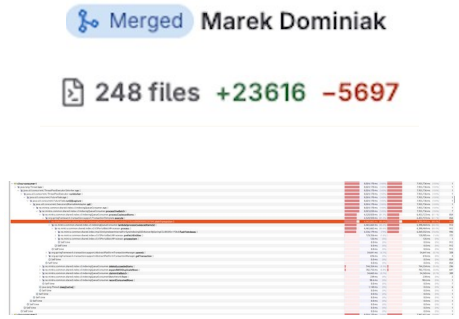
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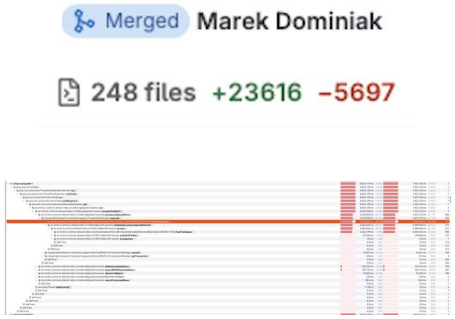
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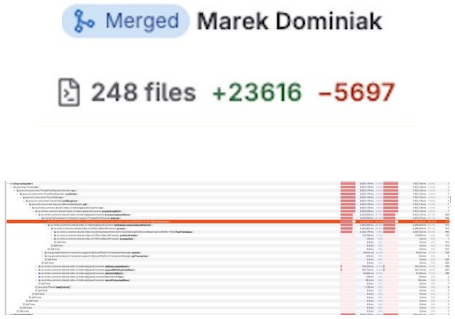
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- And shortly after **production**



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- We went to **staging**
- And shortly after **production**







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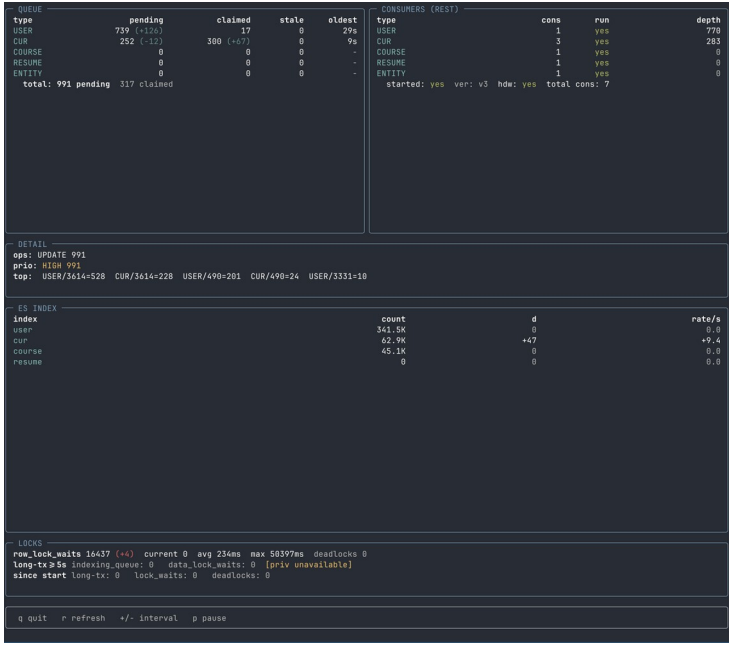
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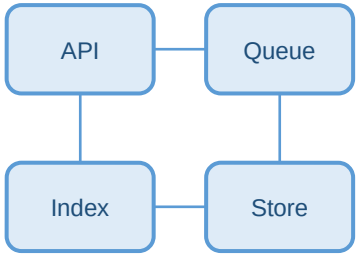
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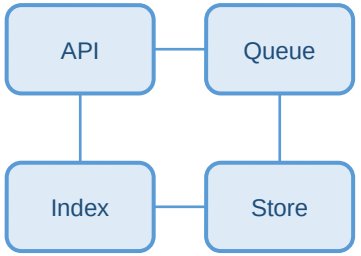
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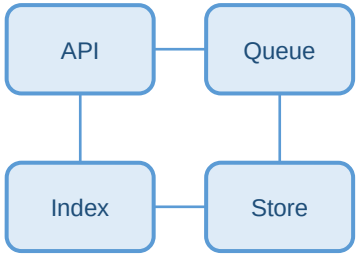
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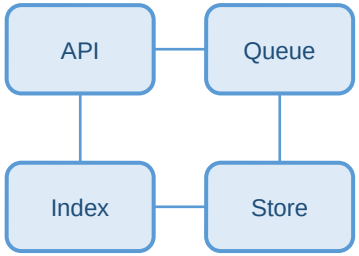
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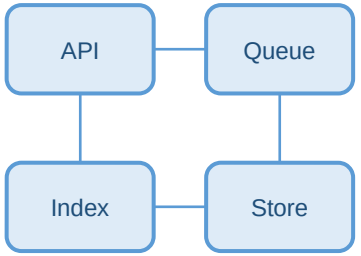
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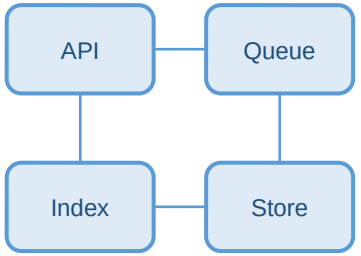
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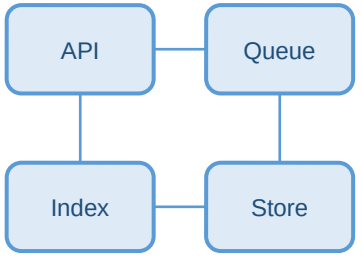
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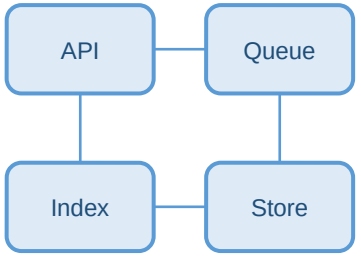
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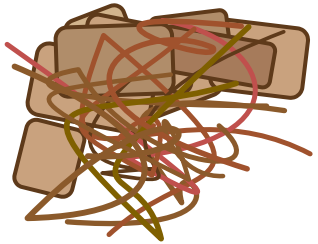
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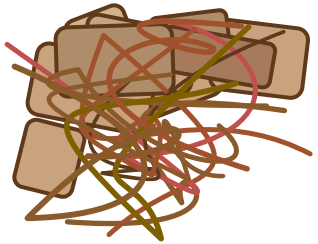


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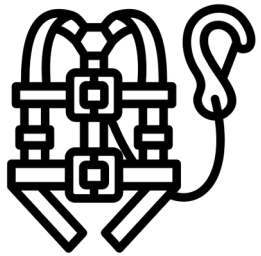
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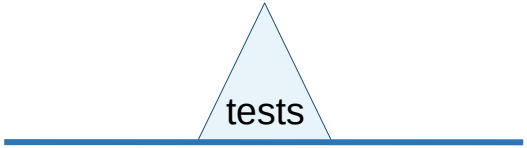
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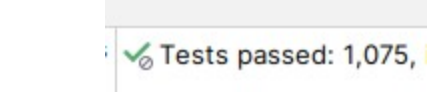
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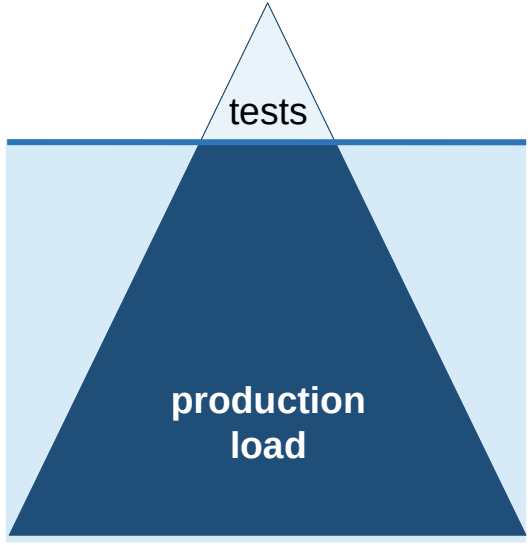
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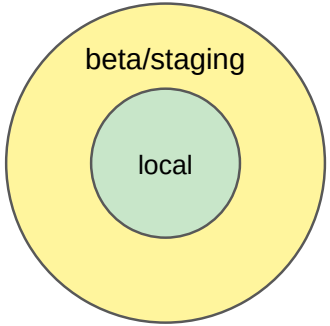
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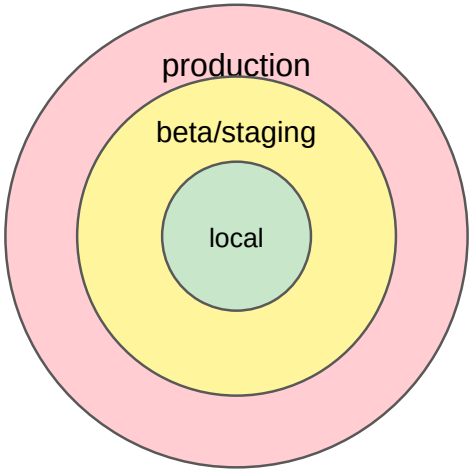
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This is what actually protects you.

- Agents can’t touch production
- Feature flags and a quick rollback
- Small MRs that someone really reads

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The safety net

	V2	V3
Feature flag	None, all or nothing	On/off per index type, Start/Stop from the dashboard
Rollback	Redeploy the previous version	Switch the flag, old path still there
Observability	General monitoring, blind to locks and queue backlog	Indexing dashboard: queue depth, throughput, success rate, failed rows, staleness alerts



Assume it will fail, so make failing cheap.

V3: simpler, safer, in production

Three alternatives

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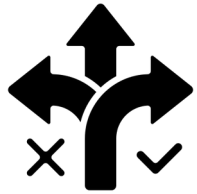
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vs



V3 implementation

- I updated the ADR. V3 MUST BE:
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- One plan caught my eye: claim-check pattern, many small steps, simpler overall



vs



V3 was simpler, not simple

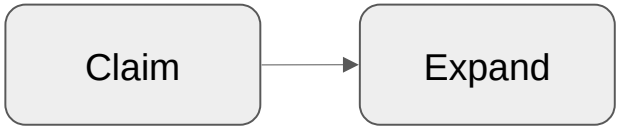
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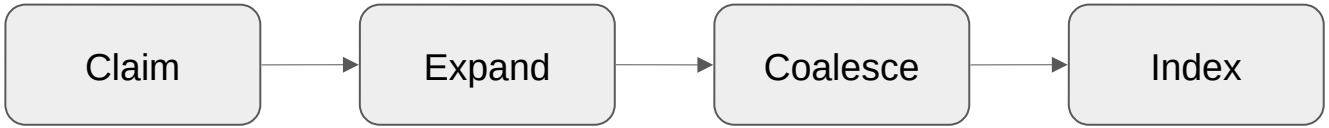
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Bulletproofing V3

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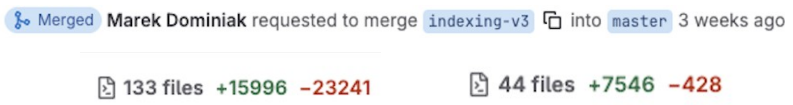
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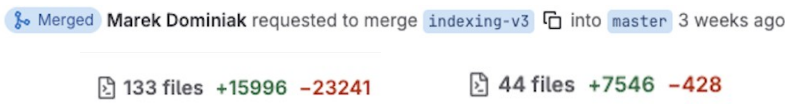
133 files +15996 −23241

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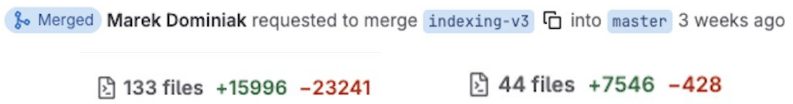
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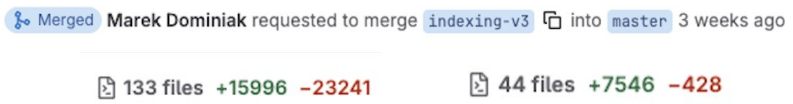
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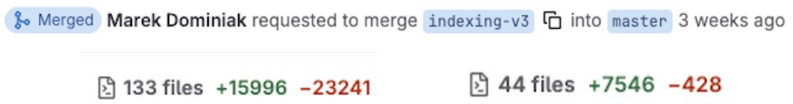
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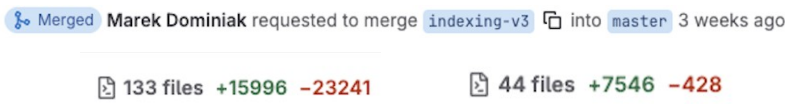
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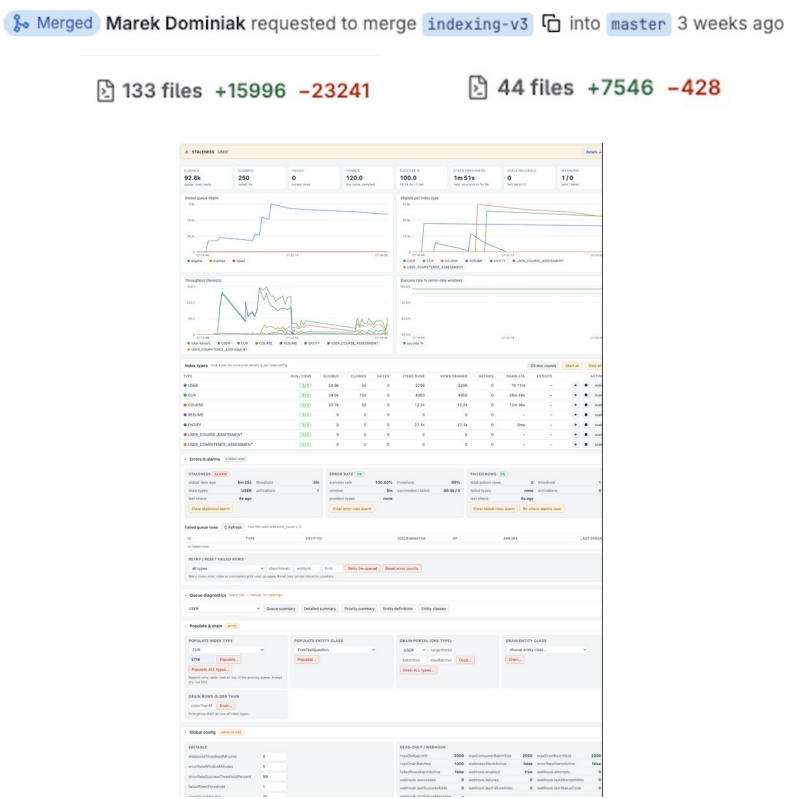
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Release expectation

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On 29 July we turn on the flag and we expect

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In the worst case, we have a very quick rollback option

Three months have passed ….

V3 in production

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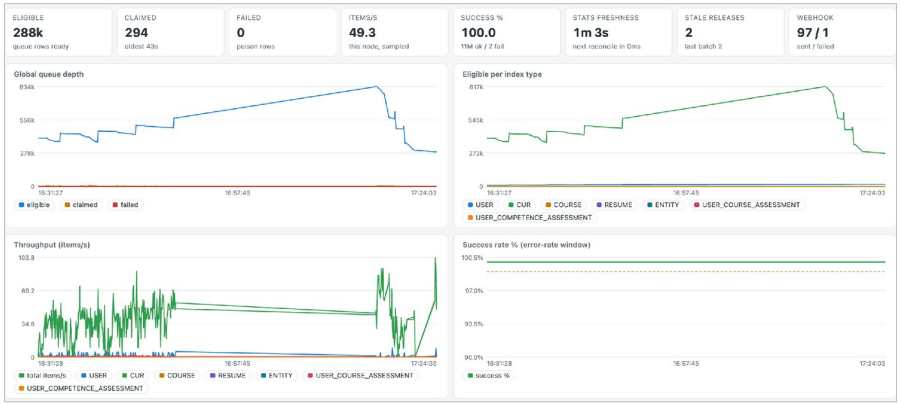
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What production looks like

Observability!

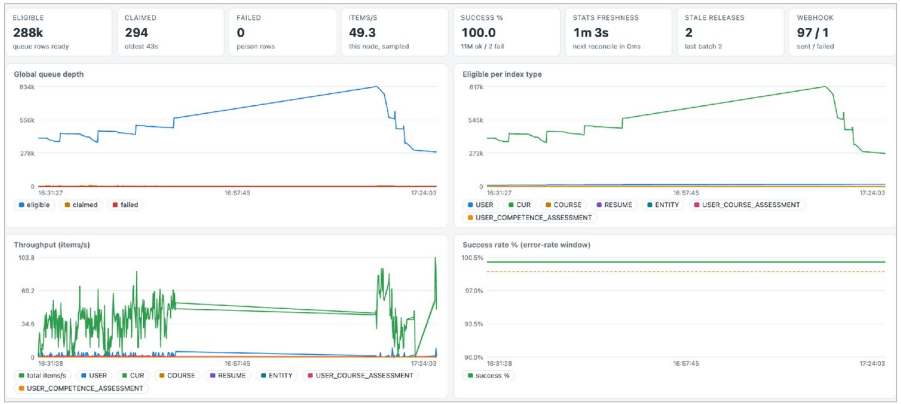
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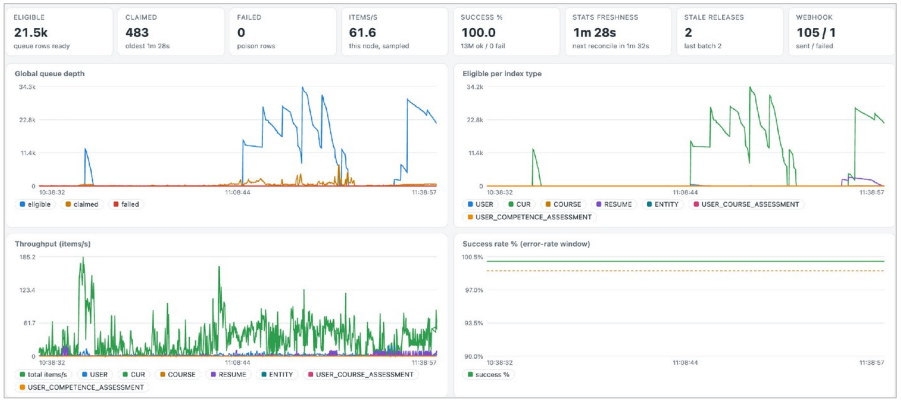
Draining a backlog of ~800k items

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What production looks like



Draining a backlog of ~800k items



Steady state: 100% success, 0 failed rows

Observability!

From V1 to V3

Each version learned from the last. The one real mistake: no way back for V2.

From V1 to V3

V1 · 14 years old

DB queue, Quartz, JMS, listeners

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Same goals, much simpler

Small transactions, claim-check

Behind a flag, V1 kept as fallback

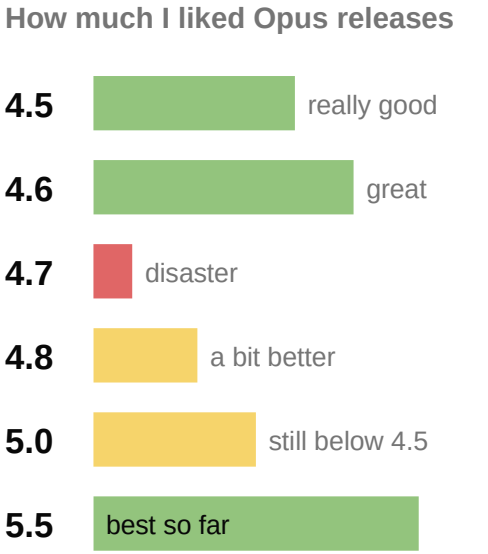
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Lessons Learned

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Don't build your whole process around one magical model / tool.

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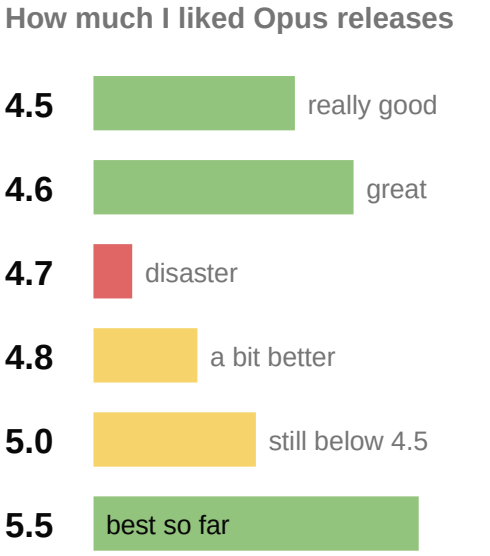


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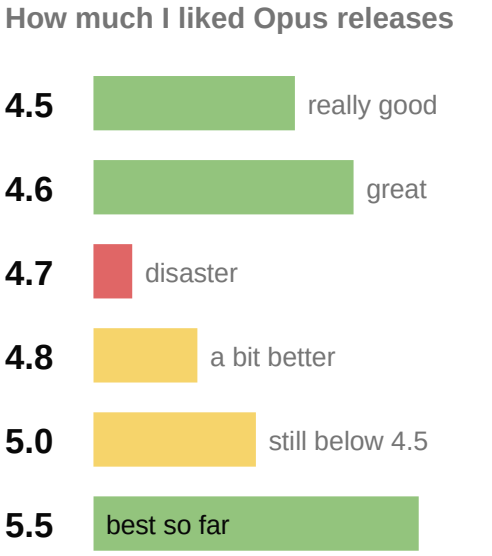


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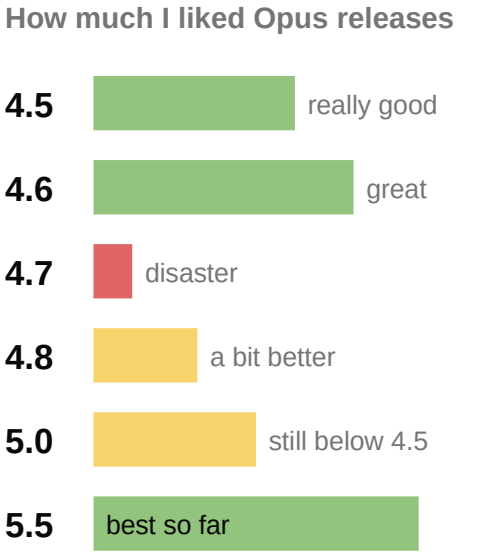


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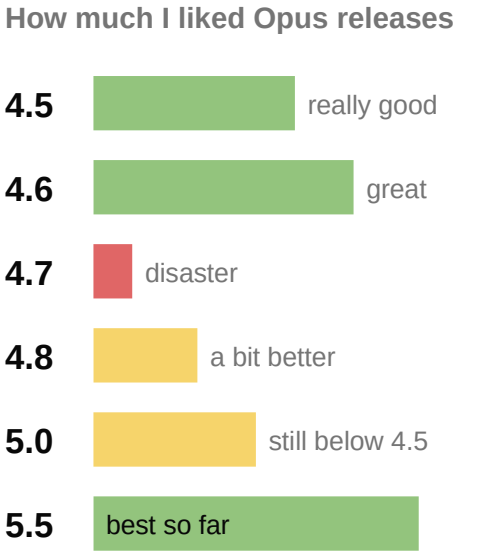


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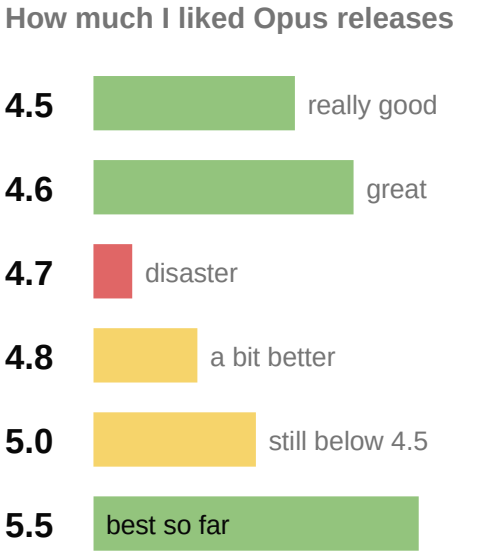


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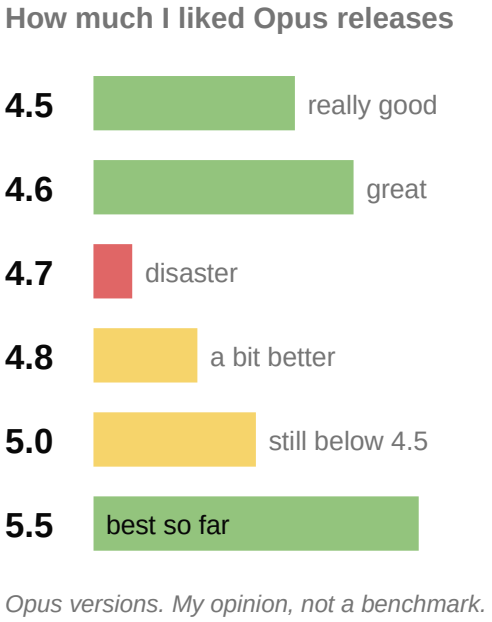


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Treat tokens as a scarce resource



Plan for tokens the way you plan for CPU and memory.

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- **Engineer around it:** small sessions, cheaper models for simple tasks, reusable plans and skills



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After half the context, quality drops.

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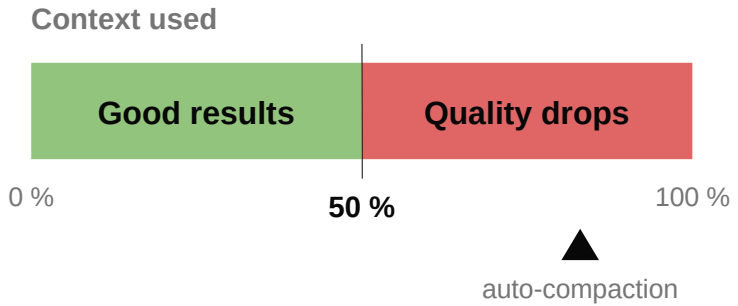
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Listen to your code reviews

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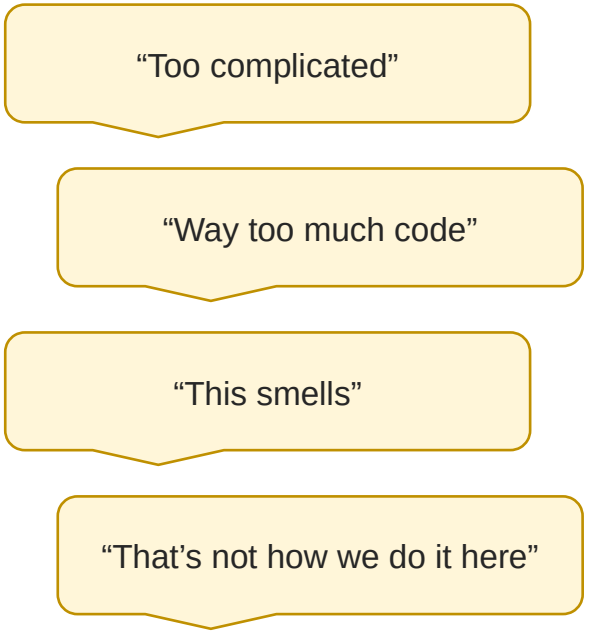
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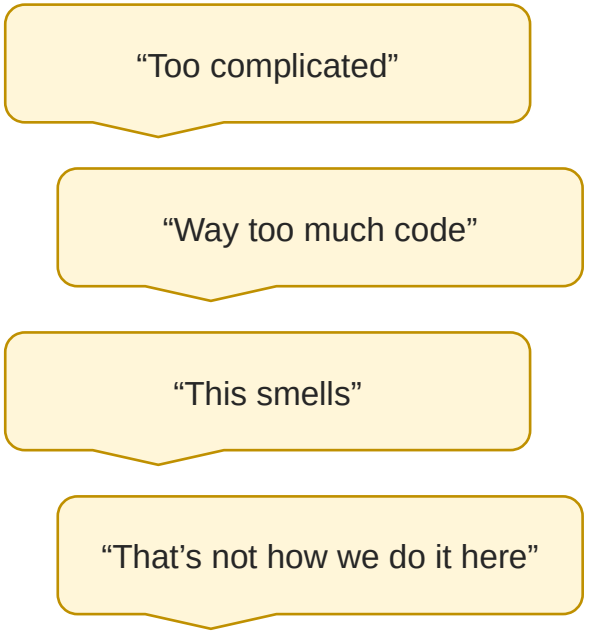
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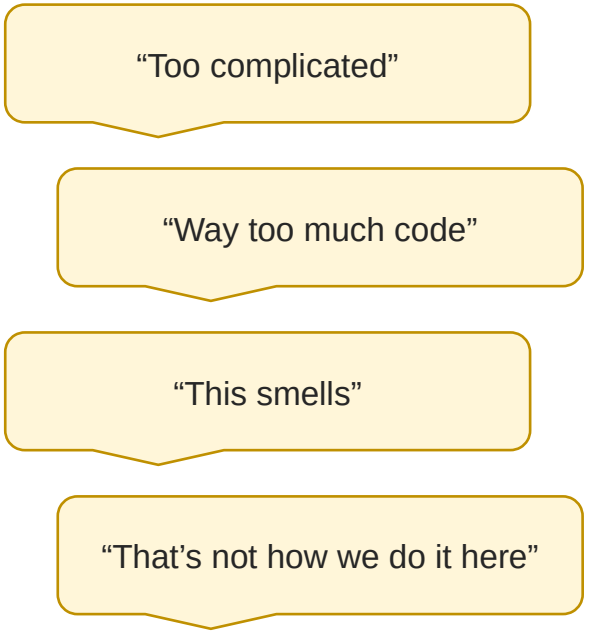
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- Our rule now: you must understand and explain every line of your MR



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Teach the agent once: rules and skills

CLAUDE.md / AGENTS.md: read in every session

```
### Comments
- Explain why, not what.
- Keep them short.

## Minimizing Code Growth
- Before creating a new class, look for an
  existing one you can extend or reuse.

## Reusing Existing Integration Tests
- Prefer a new test method in an existing class.
- No new @ContextConfiguration or @DirtiesContext:
  each forces a second Spring context.
```

Skills: team know-how it can run

/migration-risk

Which Liquibase changesets can lock hot tables, and which can run by hand before the deploy. Used for production releases since March.

/idempotent-db-change

Writes an index or data change guarded by preConditions (MARK_RAN), then proves it on a scratch schema.

Rules and skills live in the repo: peer-reviewed like code.

Agentic AI is a double-edged sword

Keep a person in the loop every few iterations.

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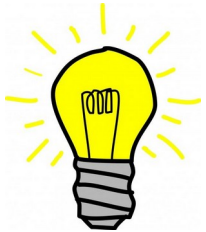
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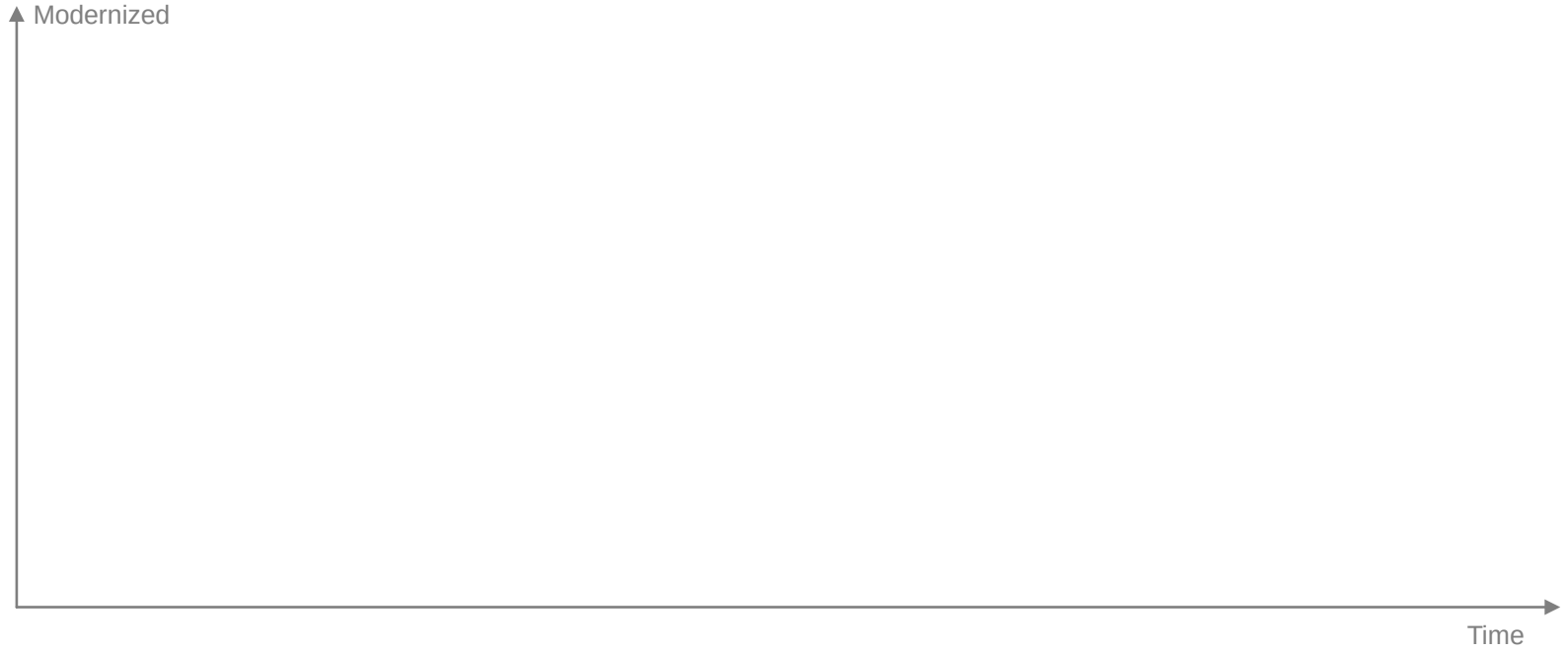
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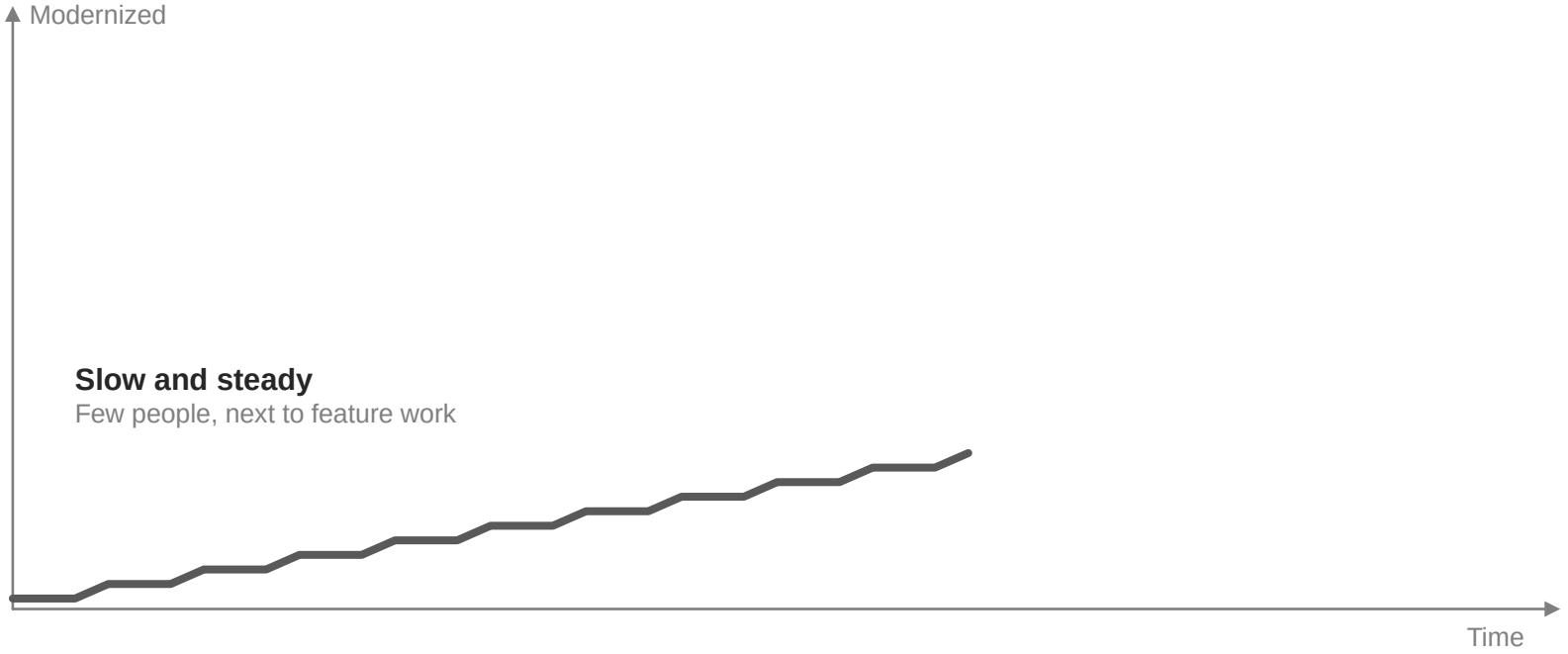
The light at the end of the tunnel



Illustrative, not measured.

For years we couldn’t say when we’d be done. Now we can see it.

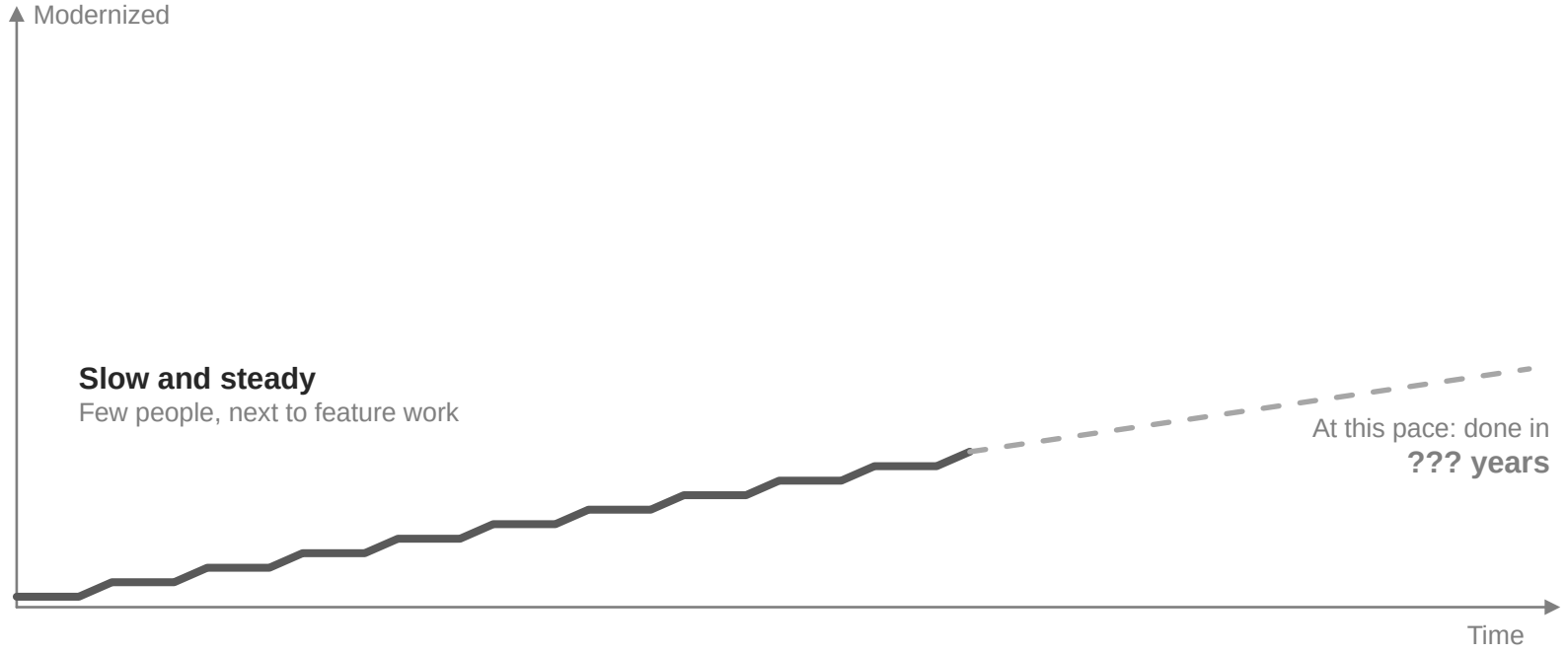
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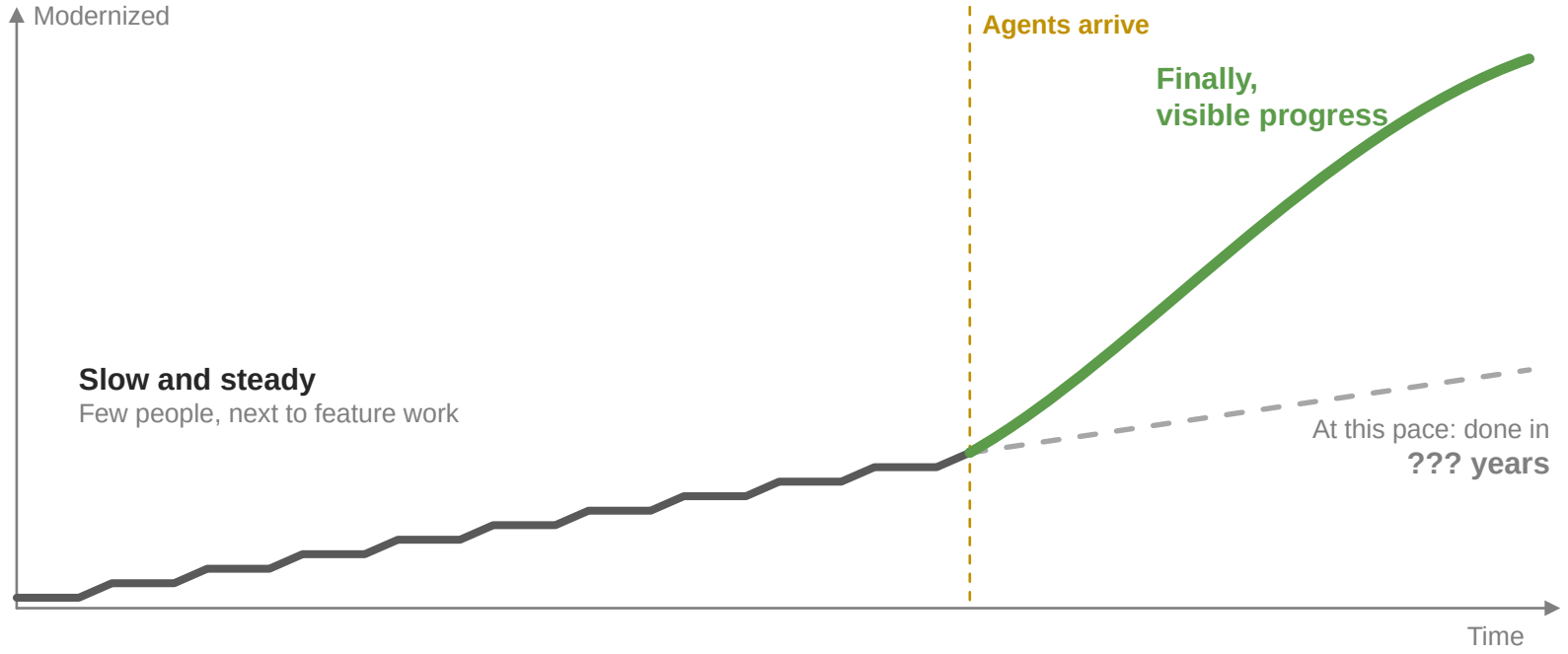
The light at the end of the tunnel



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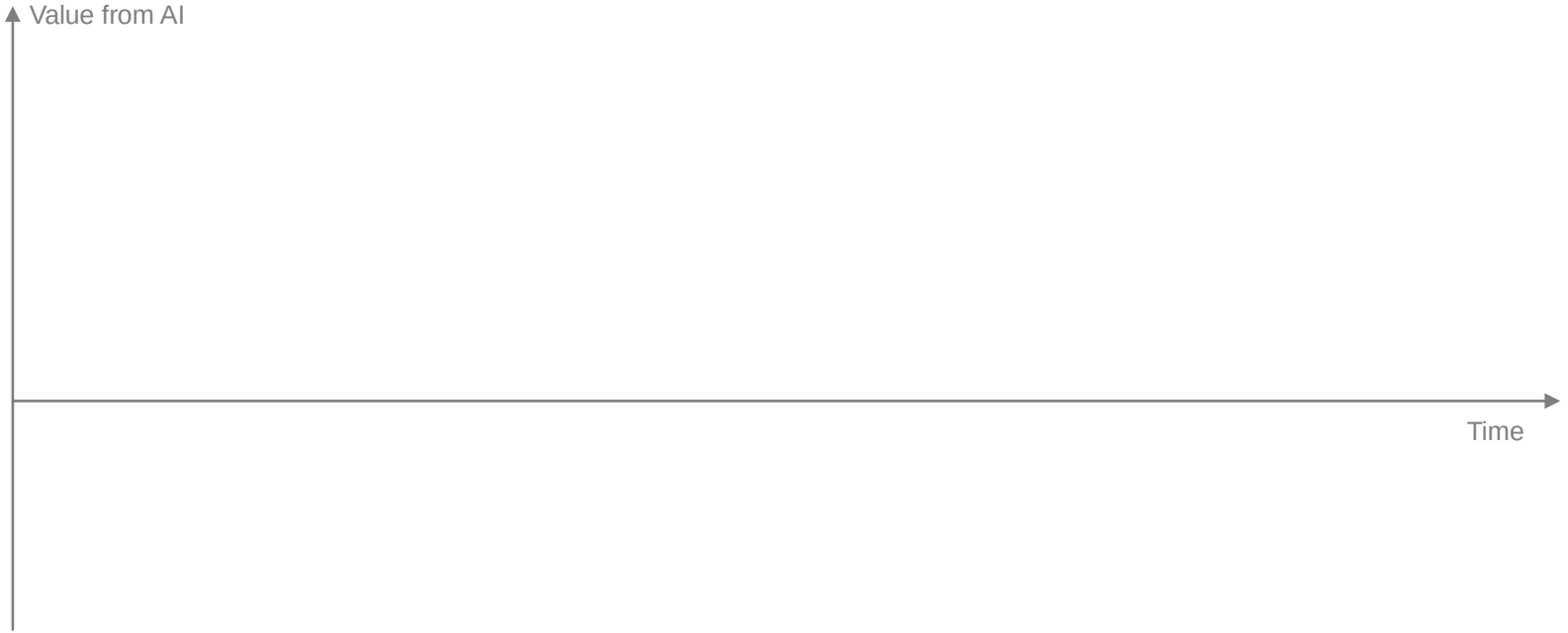
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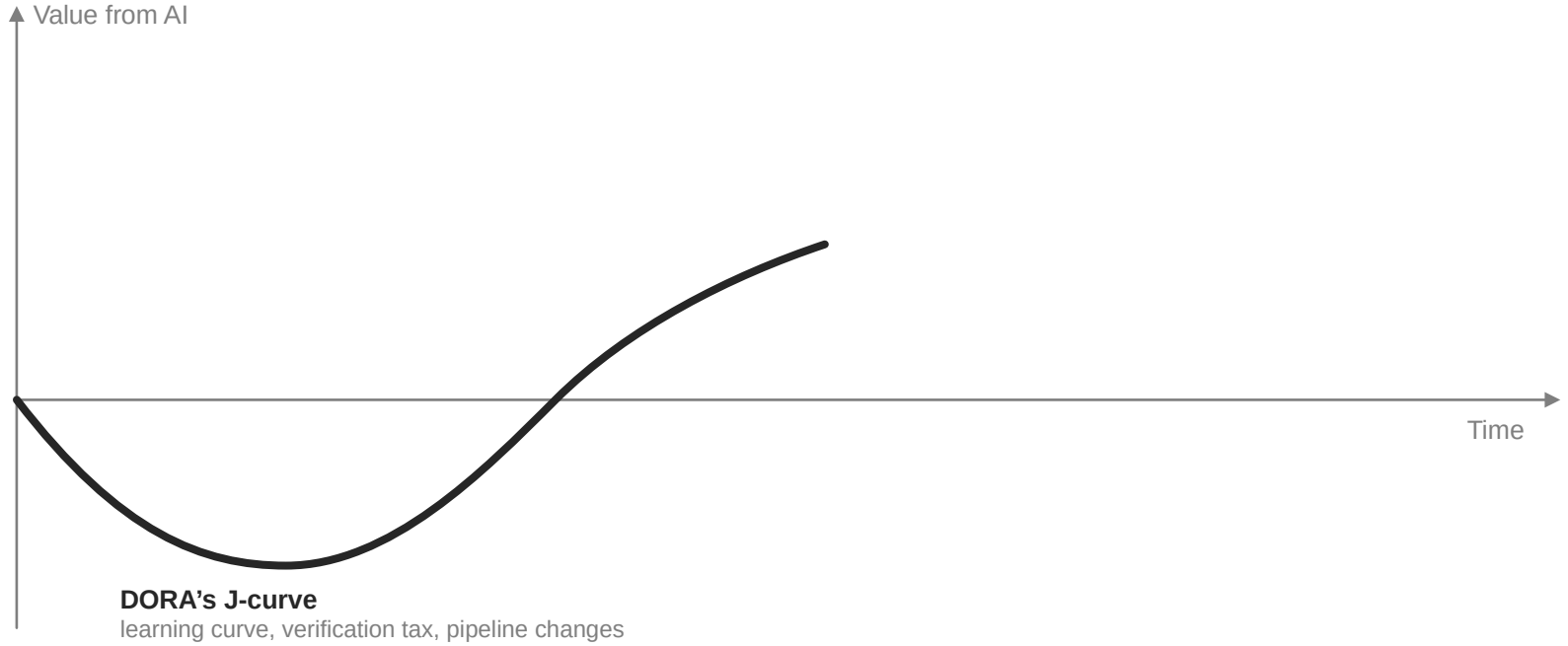
The verification tax



Sources: DORA, State of AI-assisted Software Development (2025); ROI of AI-Assisted Software Development (2026).

Code got cheap to write. Checking it and owning it still takes time.

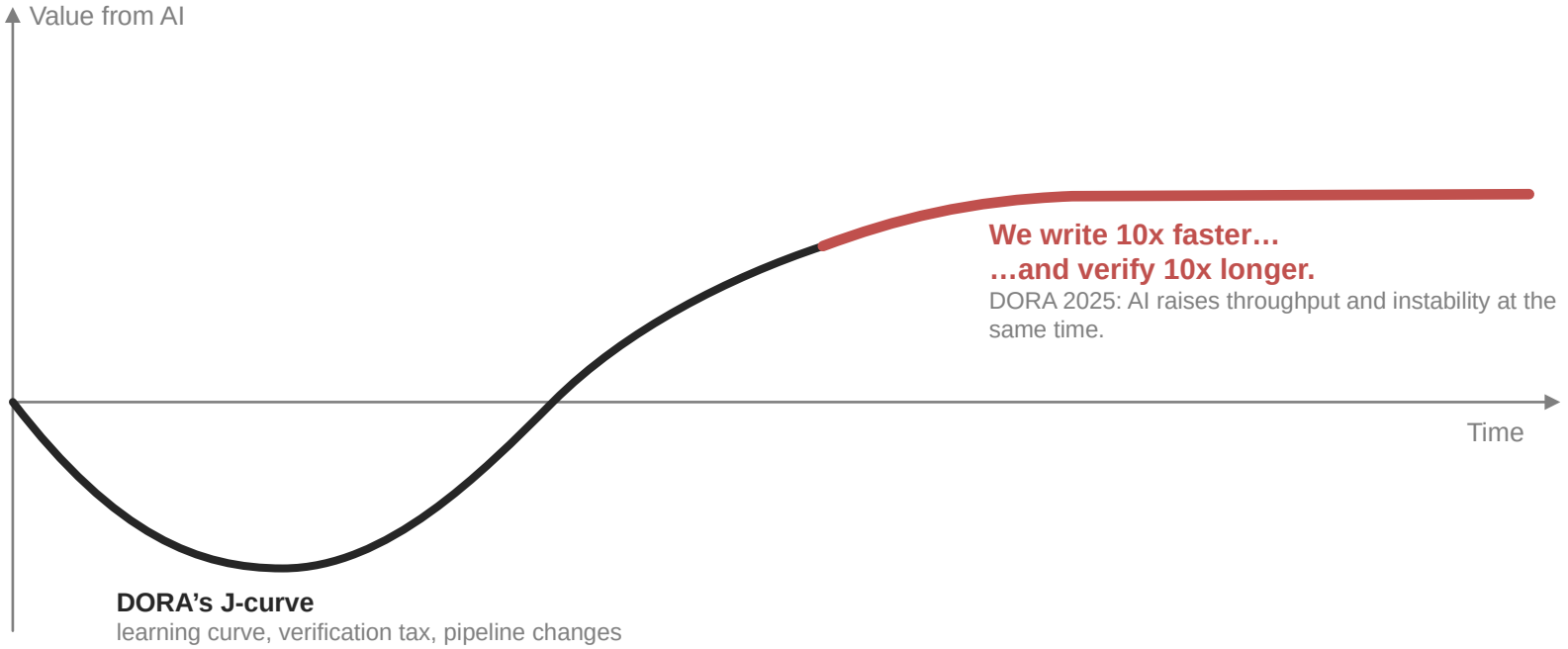
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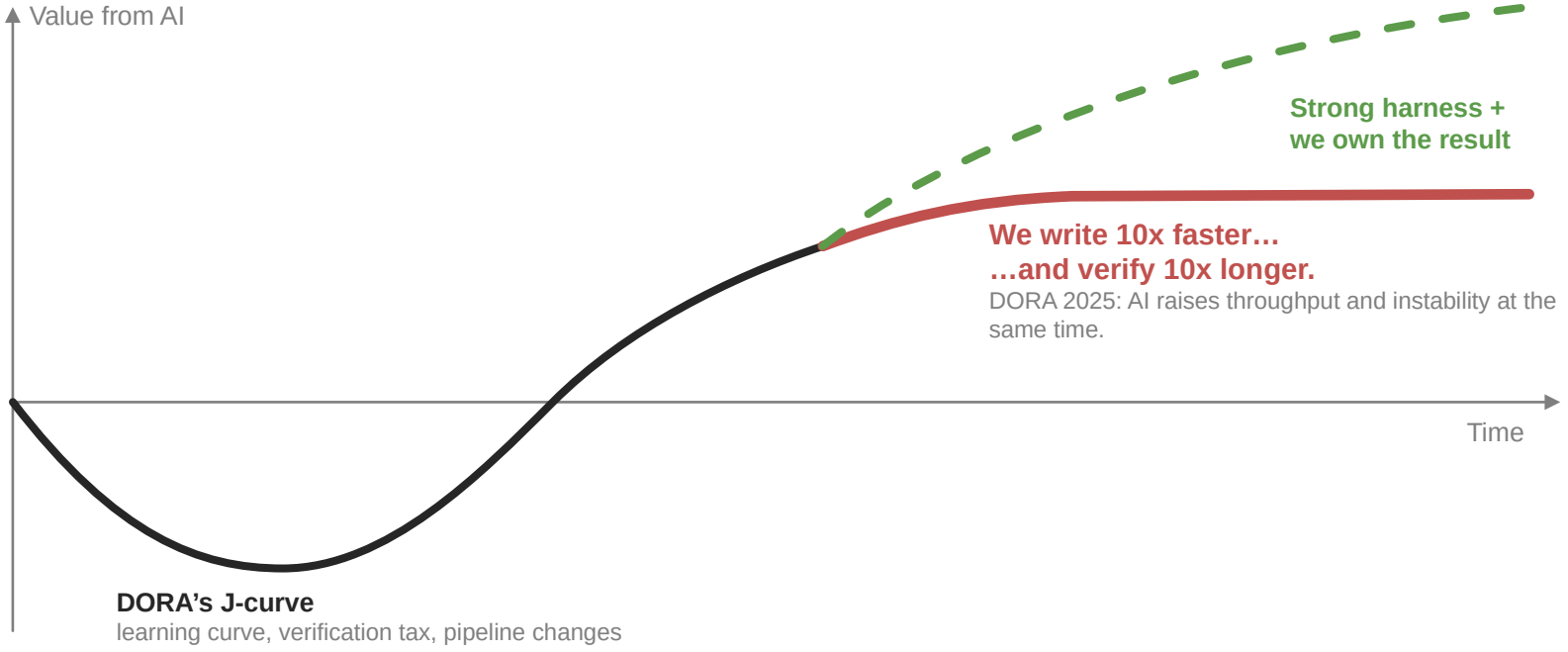
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Code got cheap to write. Checking it and owning it still takes time.

You get paid to think.

Don’t be a meat proxy.

You get paid to think.

The agent writes the code. You decide what it should do, challenge it, and own the result.

Don’t be a meat proxy.

Q/A

trainitek



marekdominiak

Presentation Slides